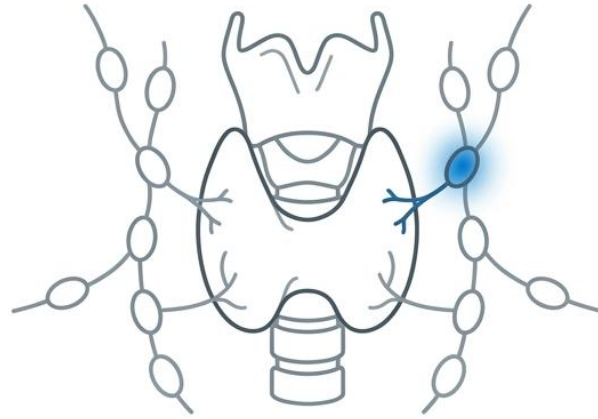


# Advances in Thyroid Cancer Diagnosis: Fluid Biomarkers



## Redefining Lymph Node Diagnosis in Thyroid Cancer

Overcoming the limits of Washout Tg with CYFRA 21-1

가톨릭의대 서울성모병원 내분비내과

임 동 준

# Today's Talk

- Standard of diagnosis of lymph node metastasis in thyroid cancer  
→ **Fine needle aspiration cytology (FNAC) + FNA washout thyroglobulin**  
**& unmet needs for proper diagnosis**
- Cytokeratin 19 & CYFRA 21-1 as an adjunctive FNA washout biomarker  
in thyroid cancer LN metastasis  
→ From biomarker mining to clinical application

# Lymph Nodes Metastases in Thyroid cancer

- Nodal metastases in thyroid cancer :

- Detected clinically in 20–31% of cases at the time of preoperative evaluation
- Local recurrence : 5-20 % of cases during cancer surveillance after initial treatment

Haugen BRM et al., 2015 American Thyroid Association Management Guidelines for Adult Patients with Thyroid Nodules and Differentiated Thyroid Cancer. Thyroid 2016; 26: 1-133.

- Accurate identification of metastatic LNS is an important issue in the management of patients with PTC to avoid unnecessary neck dissection
- Diagnostic tool for detection of metastatic LNs
  - Thyroid ultrasonography (US) , US guided fine needle aspiration cytology (FNAC) and measurement of thyroglobulin (Tg) in fine needle aspiration washout fluid

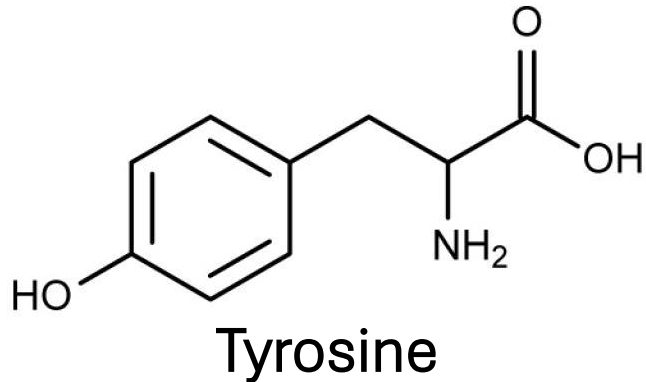
# Thyroglobulin

- 660 kDa, dimeric protein, produced by the follicular cells
- contains 48 exons
- occupies half of the protein content in thyroid

Per one thyroglobulin (Tg)

- 100~120 tyrosines
- Iodination – 25~30 tyrosines
- Finally, 10 thyroid hormone molecules per one Tg

# Synthesis of Thyroxine (T4) or Triiodothyronine (T3) from Two Iodotyrosines

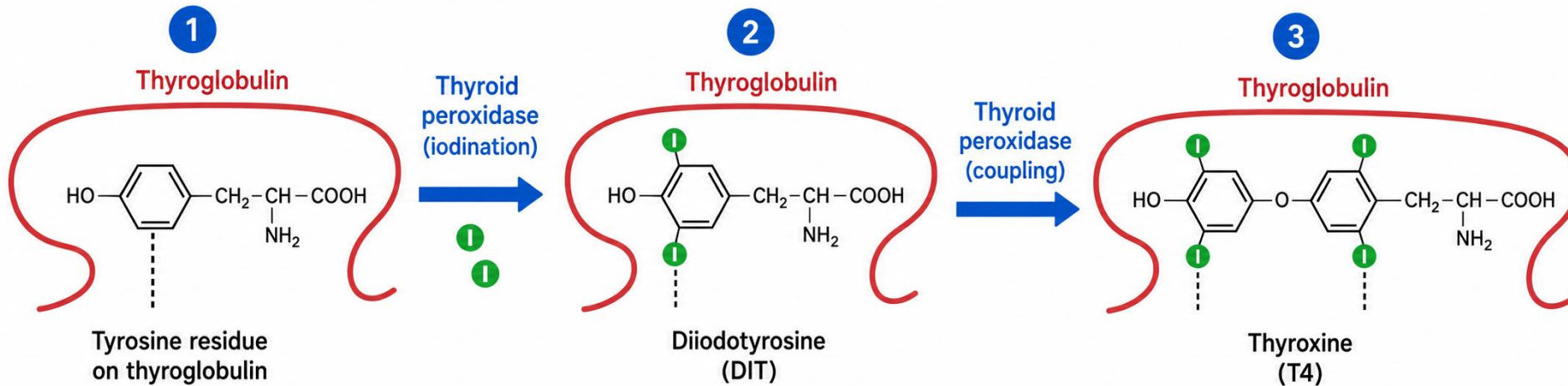


Per one thyroglobulin (Tg)

- 100~120 tyrosines
- Iodination – 25~30 tyrosines
- Finally, 10 thyroid hormone molecules per one Tg

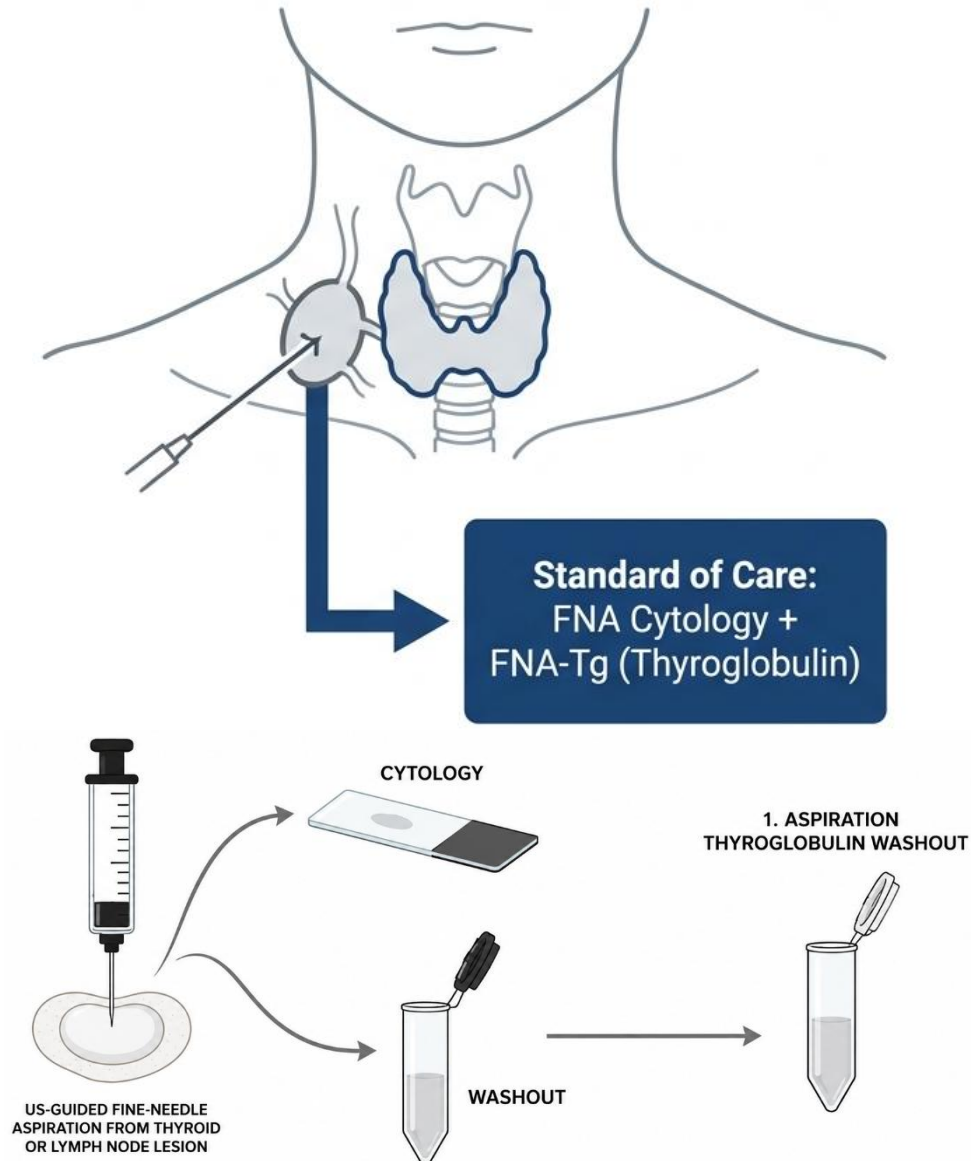
## Formation of Thyroxine (T4)

Thyroid peroxidase catalyzes iodination and coupling of tyrosine residues on thyroglobulin.



# Diagnostic Tool for Detection of Metastatic Lymph Nodes in Differentiated Thyroid Carcinoma

## The Standard Pathway



### 1. US guided fine needle aspiration cytology (FNAC)

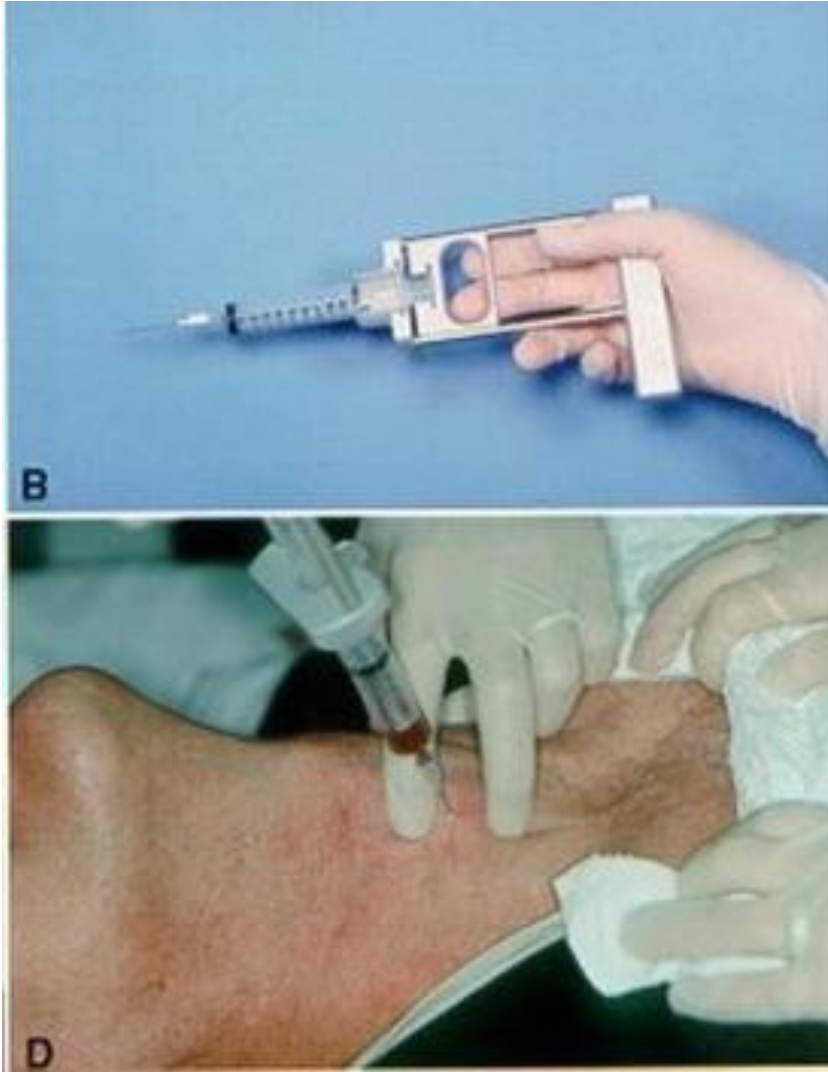
- FNAC shows disappointing sensitivity, varying from **75 to 85%** with a high rate of non-diagnostic result

### 2. Washout thyroglobulin (Tg)

- To improve the diagnostic yield of FNAC, directly **measuring the concentration of thyroglobulin (Tg) in the washout fluid** of the needle used in has been proposed.
- Thyroglobulin (Tg) : a 660 kilodalton glycoprotein that is **produced by thyroid follicular cells**, and its **appearance in non-thyroidal tissue is an evidence of recurrence or metastasis**

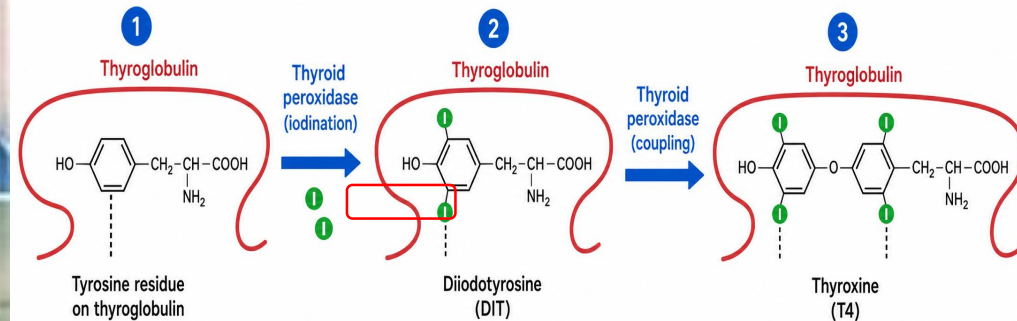
Pacini F et al., Detection of thyroglobulin in fine needle aspirates of nonthyroidal neck masses: a clue to the diagnosis of metastatic differentiated thyroid cancer. *J Clin Endo Metab.* 1992 Jun;74(6):1401-4

# 세침흡인세포검사 (Fine needle aspiration cytology, FNAC)



## Formation of Thyroxine (T4)

Thyroid peroxidase catalyzes iodination and coupling of tyrosine residues on thyroglobulin.



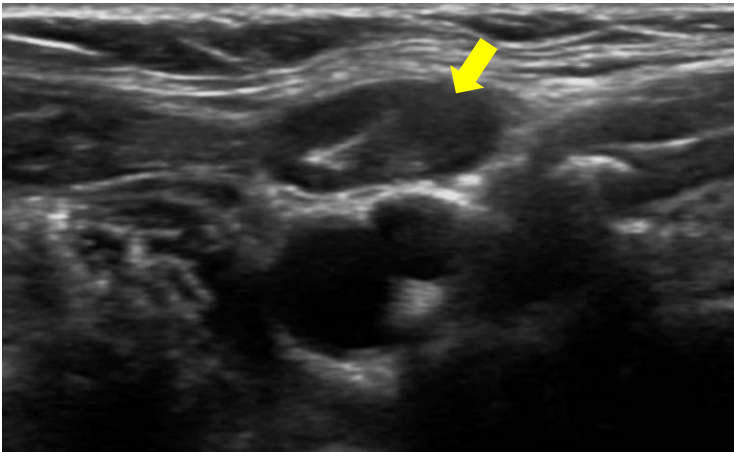
Thyroglobulin (Fine needle washout)

0.33 ng/ml (0.00~35.00) vs. **1235.4 ng/ml**

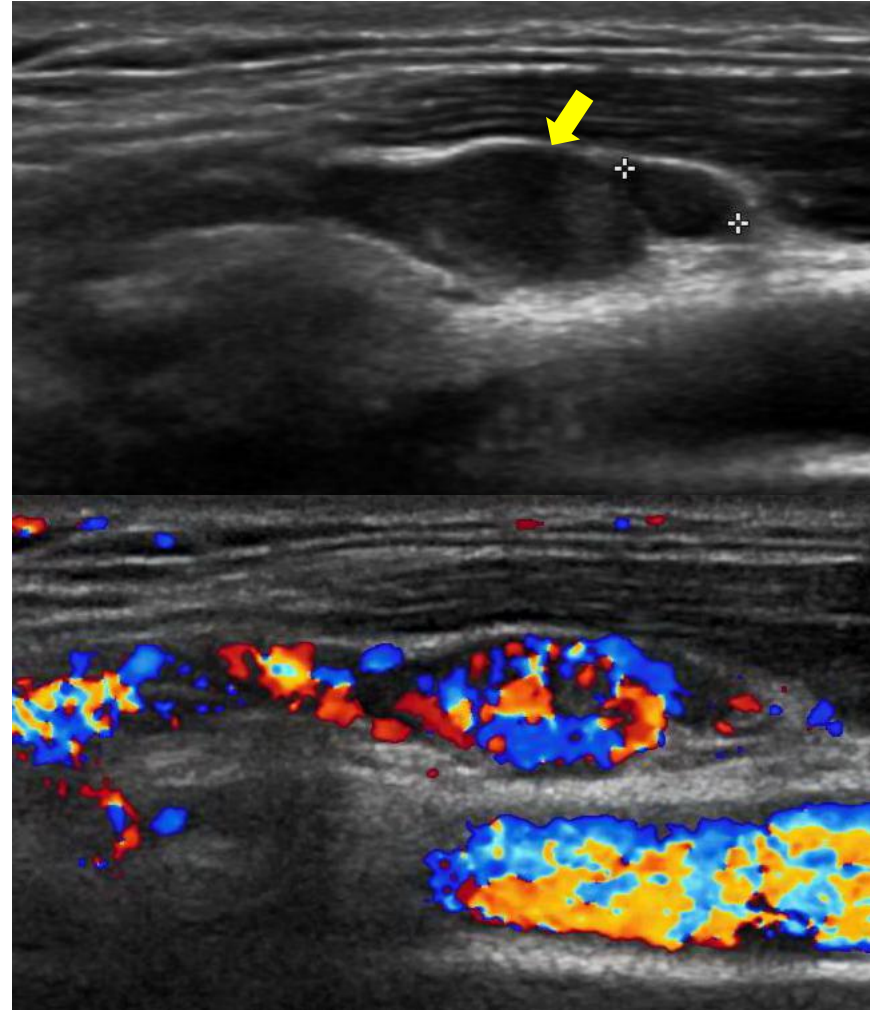
## 경부초음파에서 갑상선암 재발이 의심된 림프절



39세 남자, Lt level 2 cystic LN  
⇒ USgFNA: PTC recurrence



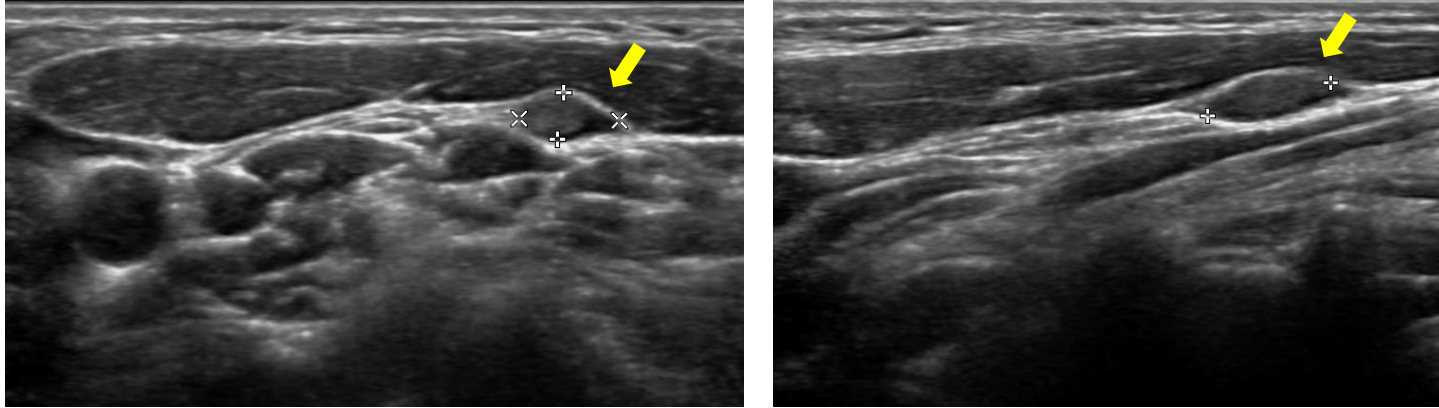
61세 여자, Rt level 3 calcified LN  
⇒ USgFNA: PTC recurrence



49세 남자, Rt level 3 LN  
⇒ USgFNA: PTC recurrence

USgFNA, Ultrasonography-guided fine needle aspiration

## A 23-year-old Female with Suspicious Lymph Node Enlargement After Total thyroidectomy and Radioactive iodine remnant ablation



A 3.5 x 6.4 x 9.0 mm sized suspicious lymph node enlargement  
R/O recurrent disease of thyroid cancer

### [DIAGNOSIS]

Lymph node, neck, LT level4

1) Adequacy of specimen

Satisfactory

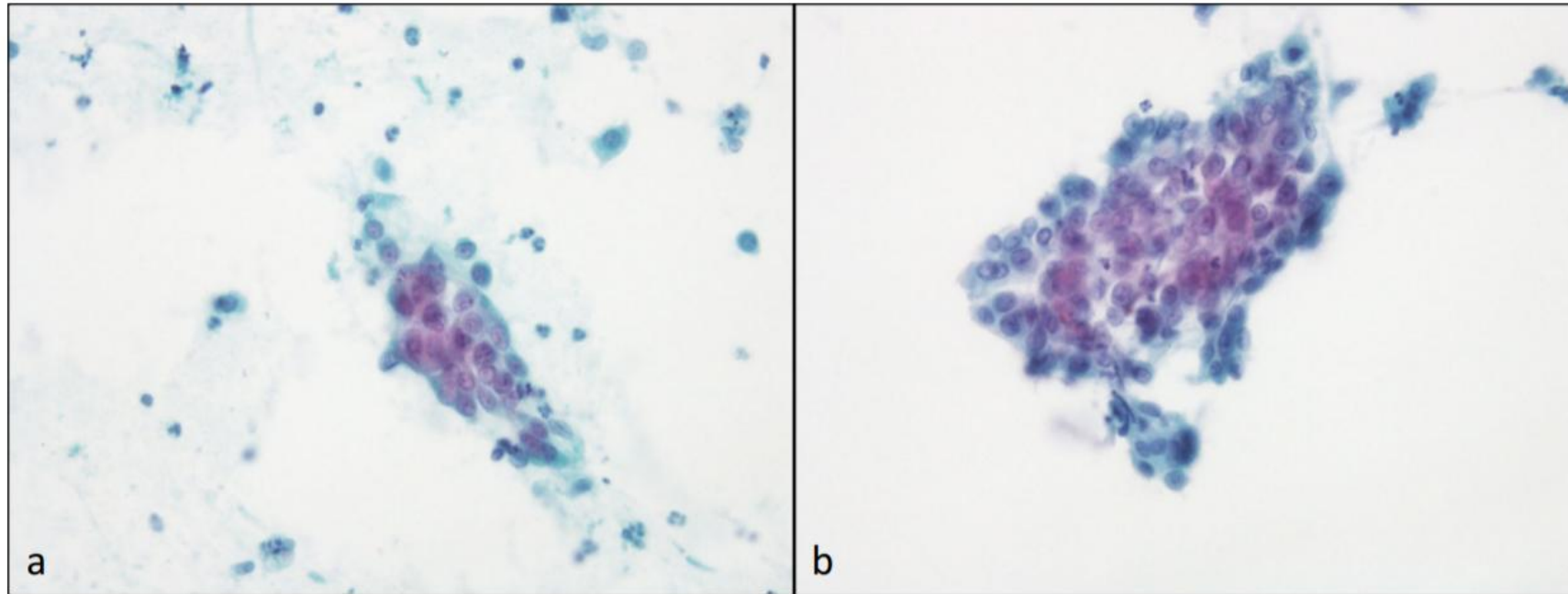
2) Cytologic diagnosis

Malignancy

Papillary thyroid carcinoma, metastatic

|                         |                |       |            |
|-------------------------|----------------|-------|------------|
| TG(Fine needle washout) | 81.73 (Lt lv4) | ng/ml | 0.00~35.00 |
|-------------------------|----------------|-------|------------|

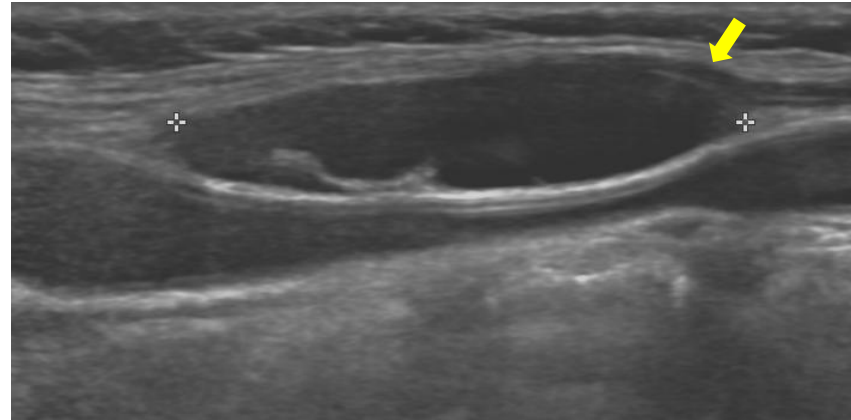
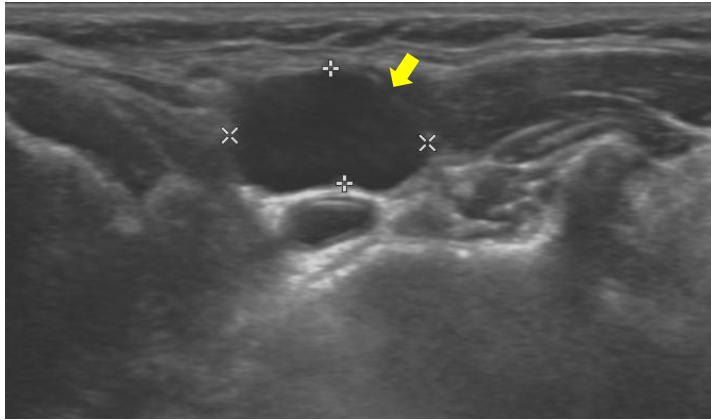
## Cervical lymph node FNA smear with Papanicolaou stain



( $\times 400$ ). Two clusters of malignant cells from a patient with history of papillary thyroid carcinoma

Diagnostic accuracy – 85%~90%

## A 39-year-old Male with Suspicious Lymph Node Enlargement After Total thyroidectomy and Radioactive iodine remnant ablation

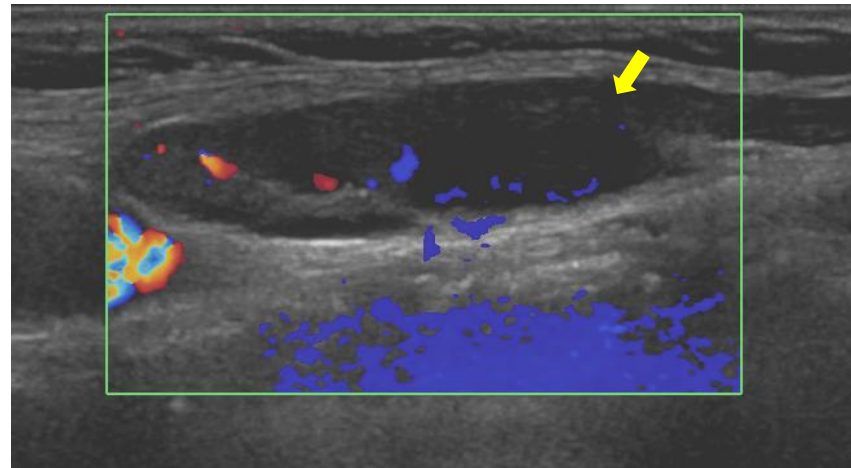


A 7 x 12 x 31 mm sized cystic mass,  
R/O recurrent disease of thyroid  
cancer

### [DIAGNOSIS]

Lymph node, neck, LT level4

- 1) Adequacy of specimen  
Not -satisfactory
- 2) Cytologic diagnosis  
cystic fluid only



TG(Fine needle washout) **350.69** (Lt lv3) ng/ml 0.00~35.00

Metastatic LN from thyroid cancer? Remnant lesion?

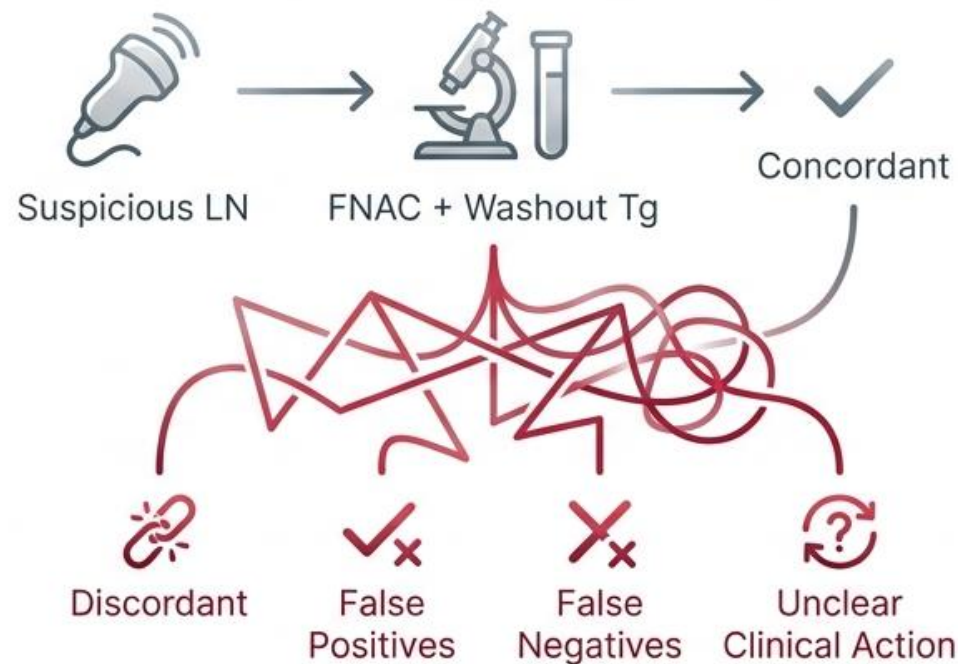
# The Gold Standard is Imperfect

FNAC combined with Washout Thyroglobulin (Tg) faces critical real-world hurdles, resulting in diagnostic ambiguity and patient risk.

## The Textbook Pathway



## The Real-World Pathway



# Cut-off Values of FNAC with Washout Tg Measurement and Diagnostic Performance According to Operative Status

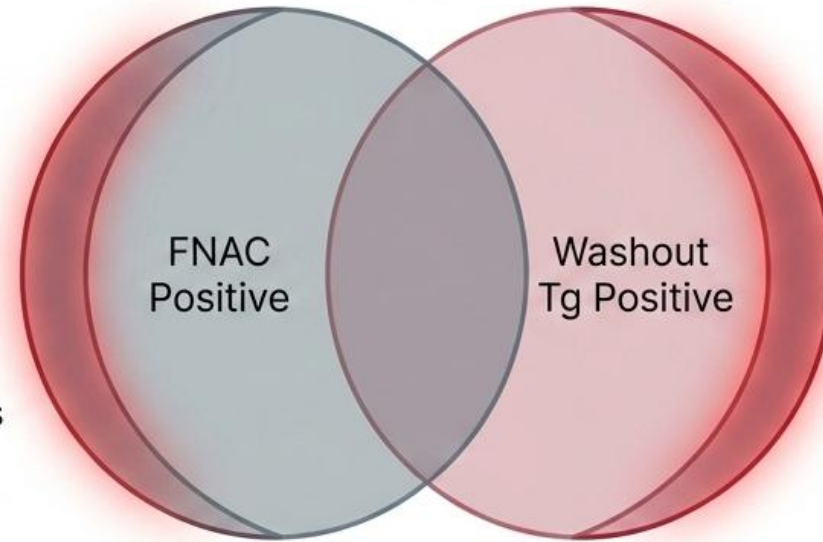
| LNs                       | Cut-off value (ng/mL) | Discordant results (%) | Sensitivity (%) | Specificity (%) | PPV (%) | NPV (%) | Accuracy (%) |
|---------------------------|-----------------------|------------------------|-----------------|-----------------|---------|---------|--------------|
| All LNs (n=707)           | 2.9                   | 15.0                   | 83.0            | 81.2            | 80.2    | 82.4    | 81.7         |
| Preoperative LNs (n=295)  | 2.9                   | 15.3                   | 87.5            | 83.4            | 69.4    | 94.0    | 84.7         |
| Preoperative LNs (n=295)  | 9.6                   | 11.5                   | 87.5            | 88.2            | 77.0    | 94.4    | 88.5         |
| Postoperative LNs (n=412) | 2.9                   | 14.8                   | 95.0            | 86.9            | 77.2    | 87.6    | 89.0         |
| Postoperative LNs (n=412) | 2.1                   | 13.3                   | 96.0            | 90.4            | 78.1    | 98.4    | 91.8         |

⇒ Due to inconclusive results, unnecessary neck dissection can occur in 13.9% (36/259 dissected cases)

# The Discordance Dilemma

**~9%**  
**Discordance**

1 in 10 patients present with conflicting FNAC and Tg results (8.9% overall discordance).



**13.9%**  
**Unnecessary Surgeries**

The clinical cost of ambiguity. Among dissected cases with inconclusive results, nearly 14% underwent unnecessary neck dissections.

Conflicting diagnostic signals force clinicians into invasive, redundant, or unnecessary surgical interventions.

# Limitation of Washout Thyroglobulin (wTg)

## 1) **Wide range of washout Tg levels and diagnostic cuff of values**

: Various Tg assays with different functional sensitivity (ranging from 0.1 to 3 ng/mL) were utilized and cutoff values ranging from 0.2 to 50 ng/mL have been suggested for washout Tg.

Giorgio G. and Angela F., Thyroglobulin in Lymph Node Fine-Needle Aspiration Washout: A Systematic Review and Meta-analysis of Diagnostic Accuracy. *J Clin Endocrinol Metab.*2014; 99(6): 1970-1985

2) **Preoperative vs postoperative assessment:** false positive washout Tg in patient with retained normal thyroid tissue after thyroidectomy or preoperative evaluation

3) **Interference of antibodies** : false negative washout Tg in patients with high thyroglobulin antibody (TgAb)

# Discordant Cases Between FNAC and Washout Tg

| No | Operation           | US finding            | FNAC                    | Washout Tg | Serum Tg | TgAb  |
|----|---------------------|-----------------------|-------------------------|------------|----------|-------|
| 1  | Lt. lobectomy       | Intermediate LN       | Thyroiditis             | 648.03     | 28.56    | 3.24  |
| 2  | Total thyroidectomy | R/O cystic metastasis | Cystic fluid only       | 902.17     | 0.09     | 8.83  |
| 3. | Total thyroidectomy | Intermediate LN       | Negative for malignancy | 15.70      | 7.57     | 52.31 |
| 4. | Total thyroidectomy | Intermediate LN       | Negative for malignancy | 120.88     | 0.62     | 67.57 |
| 5. | Total thyroidectomy | Intermediate LN       | Cystic fluid only       | 633.10     | 29.56    | 3.24  |
| 6. | Rt. lobectomy       | Suspicious LN         | Metastatic PTC          | 1.33       | 7.69     | 7.41  |
| 7. | Total thyroidectomy | Suspicious LN         | Metastatic PTC          | 0.16       | 0.21     | 8.62  |

- ✓ No clear cause for the inconsistency between the cytologic findings and washout Tg
- ✓ Few studies have been reported supporting diagnostic discrepancies

# Clinical Questions & Unmet Needs

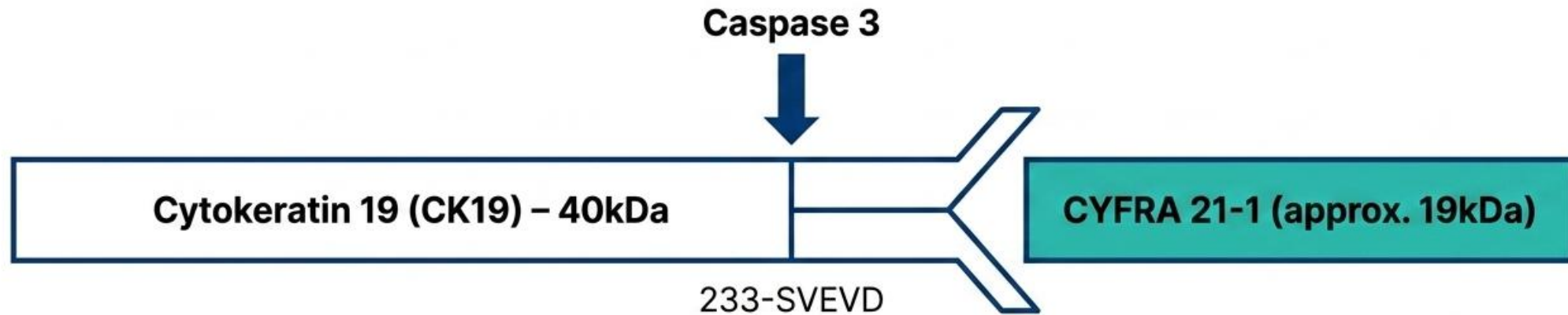
- Is there any backup biomarker to support diagnosis of recurrent/persistent metastatic LN above FNAC and/or FNA washout Tg?
- New biomarker would be
  - cancer marker, not thyroid tissue marker
  - easy to measure like thyroglobulin
  - only expressed in thyroid cancer, not in other cancer

# Objective

- Routine cytological investigation is performed mainly on the cellular component of the FNAC specimen, and less attention is given to the fluid component.
- Aim of the study :  
to seek complementary and reliable biomarker that significantly improves the accuracy of preoperative or postoperative metastatic LNs in inconsistent results

# The Molecular Origin: A Product of Tumor Apoptosis

**Cytokeratin 19 (CK19)** forms the **cytoskeleton** of **epithelial cells**. During **active tumor apoptosis**, the protease **Caspase-3** specifically **cleaves** CK19 at the 233-SVEVD site, releasing a **soluble 19kDa fragment**—**CYFRA 21-1**—into the extracellular space.

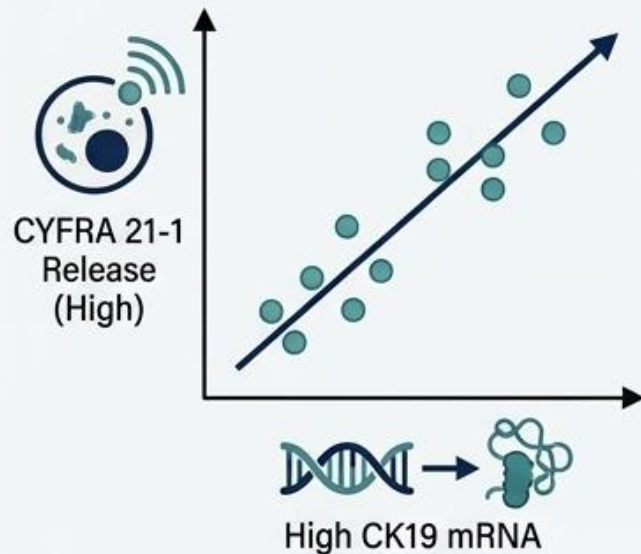


**Key Insight:** Elevated CYFRA 21-1 levels reflect not just tumor mass, but active, ongoing tumor cell apoptosis.

# Proving the Mechanism in Lung Cancer

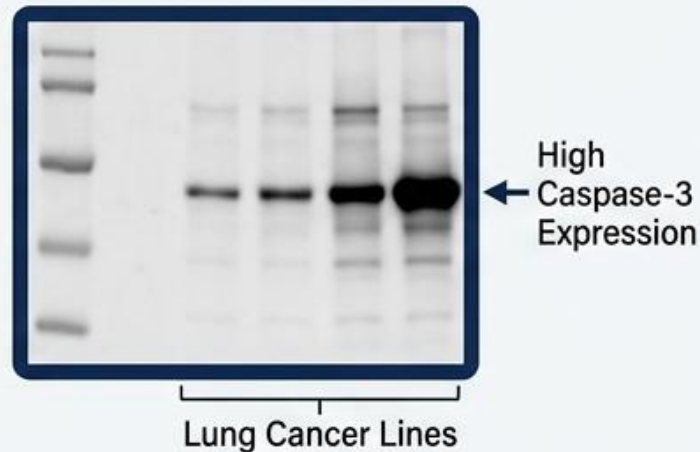
Initial studies across 13 cell lines confirmed the direct biological pathway. Cell lines releasing significant amounts of CYFRA 21-1 simultaneously exhibited high CK19 mRNA and Caspase-3 expression.

## 1. Correlation: mRNA & CYFRA 21-1 Release



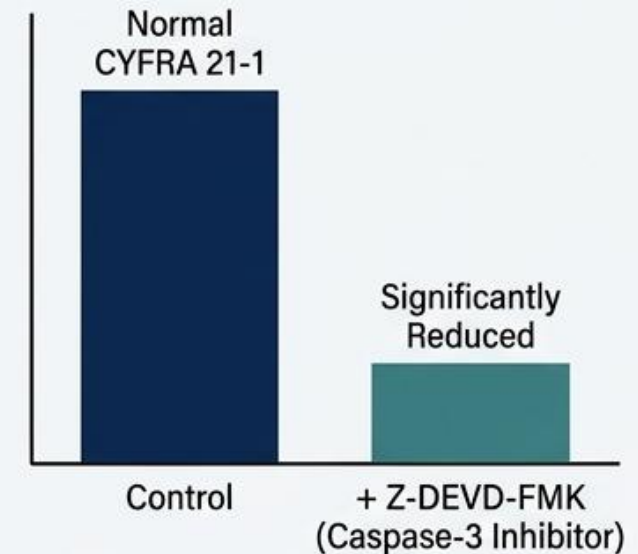
Validated across 13 lung cancer lines

## 2. Enzyme Expression: Caspase-3 Presence



Strong expression in CYFRA 21-1 releasing lines

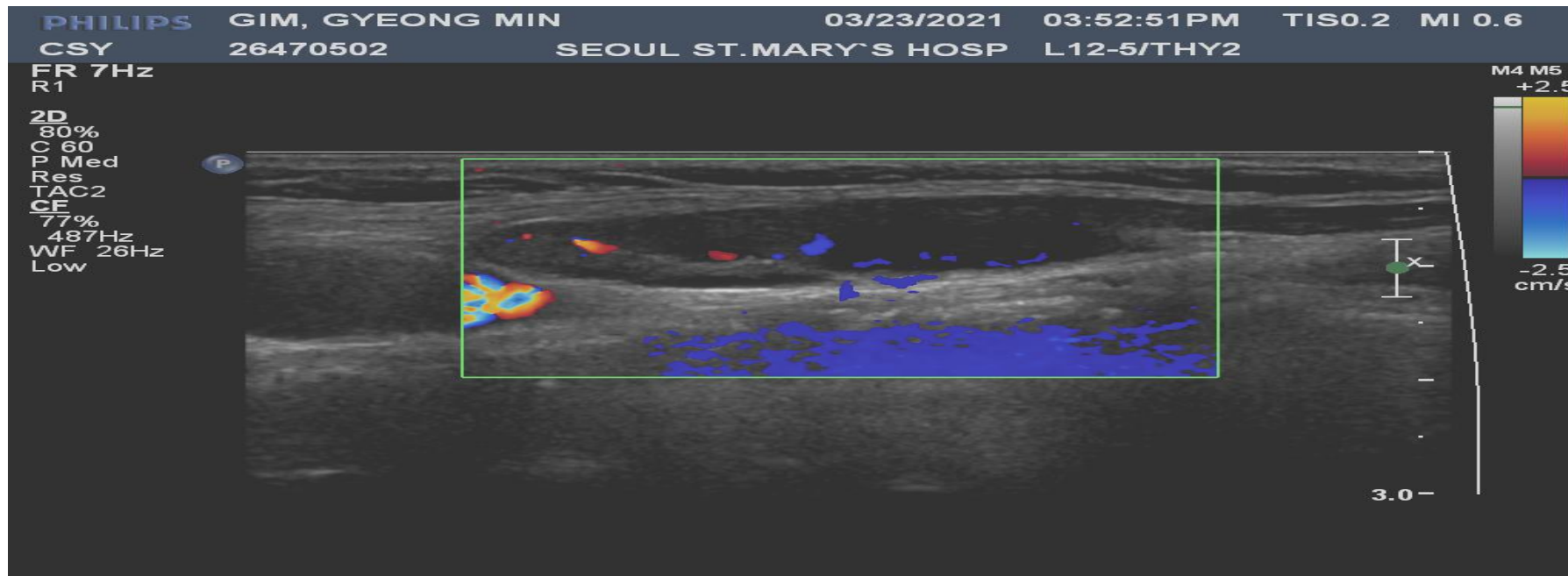
## 3. Inhibition: Halting CYFRA 21-1 Production



Inhibition halts release

# Tumor Marker of Fine-needle Aspiration Washout in Other Cancers

- Despite harboring various products of the epithelial cells' activity, the liquid component of a thyroid LN is routinely discarded in FNAC workflow. By contrast, in breast cancer and lung cancer, the protein content of fluids is frequently investigated regarding their protein content.
- Recent studies have suggested that **measurement of tumor marker concentrations of body tissue or fluid of metastatic LNs** in patients with known malignancy could be helpful to diagnose the presence of tumors.



Suh YJ et al., Tumor markers in fine-needle aspiration washout for cervical lymphadenopathy in patients with known malignancy: preliminary study. *Am J Roentgenol*. 2011 Oct;197(4):w730-6

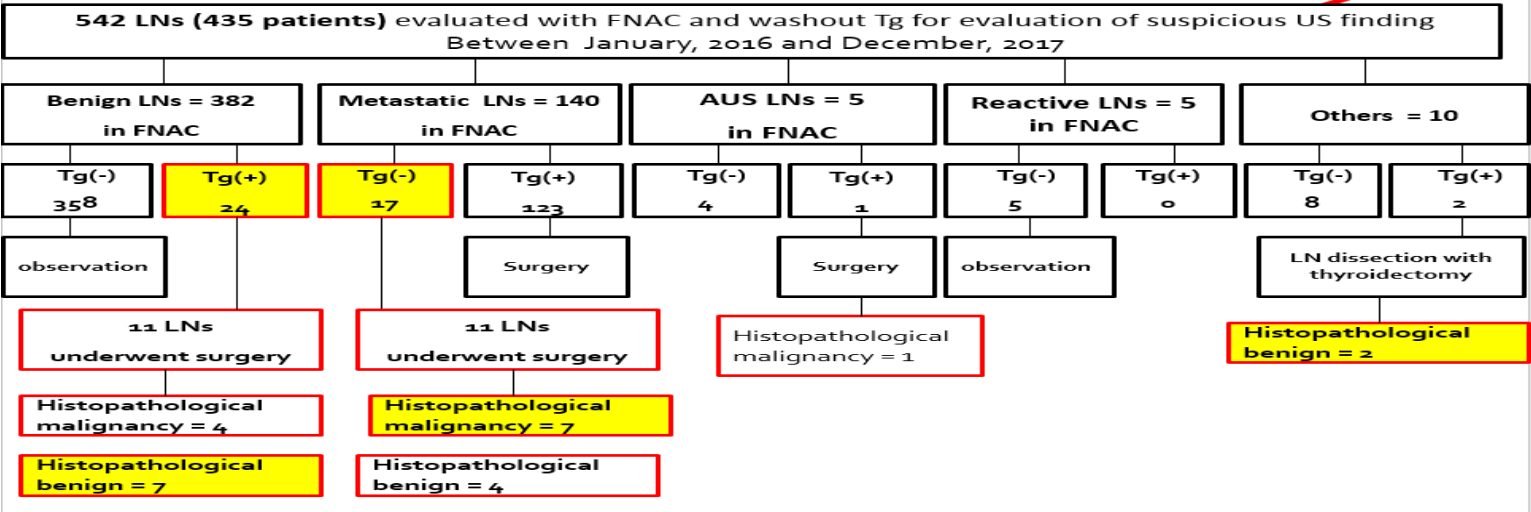
# CYFRA 21-1 Dramatically Outperforms Traditional Markers

| Comparative Superiority Matrix |              |              |
|--------------------------------|--------------|--------------|
| Method                         | Sensitivity  | Accuracy     |
| US-FNAB Alone                  | 79.0%        | 89.1%        |
| + CEA                          | 43.2%        | 69.9%        |
| + CA15-3                       | 59.3%        | 70.5%        |
| <b>+ CYFRA 21-1</b> ✓          | <b>91.4%</b> | <b>93.6%</b> |

**The Clinical Impact:** Adding CYFRA 21-1 to the diagnostic protocol significantly improves Negative Predictive Value (NPV) and overall accuracy. This simple fluid analysis has the potential to prevent up to 75% of unnecessary invasive intraoperative Sentinel Node Biopsies.

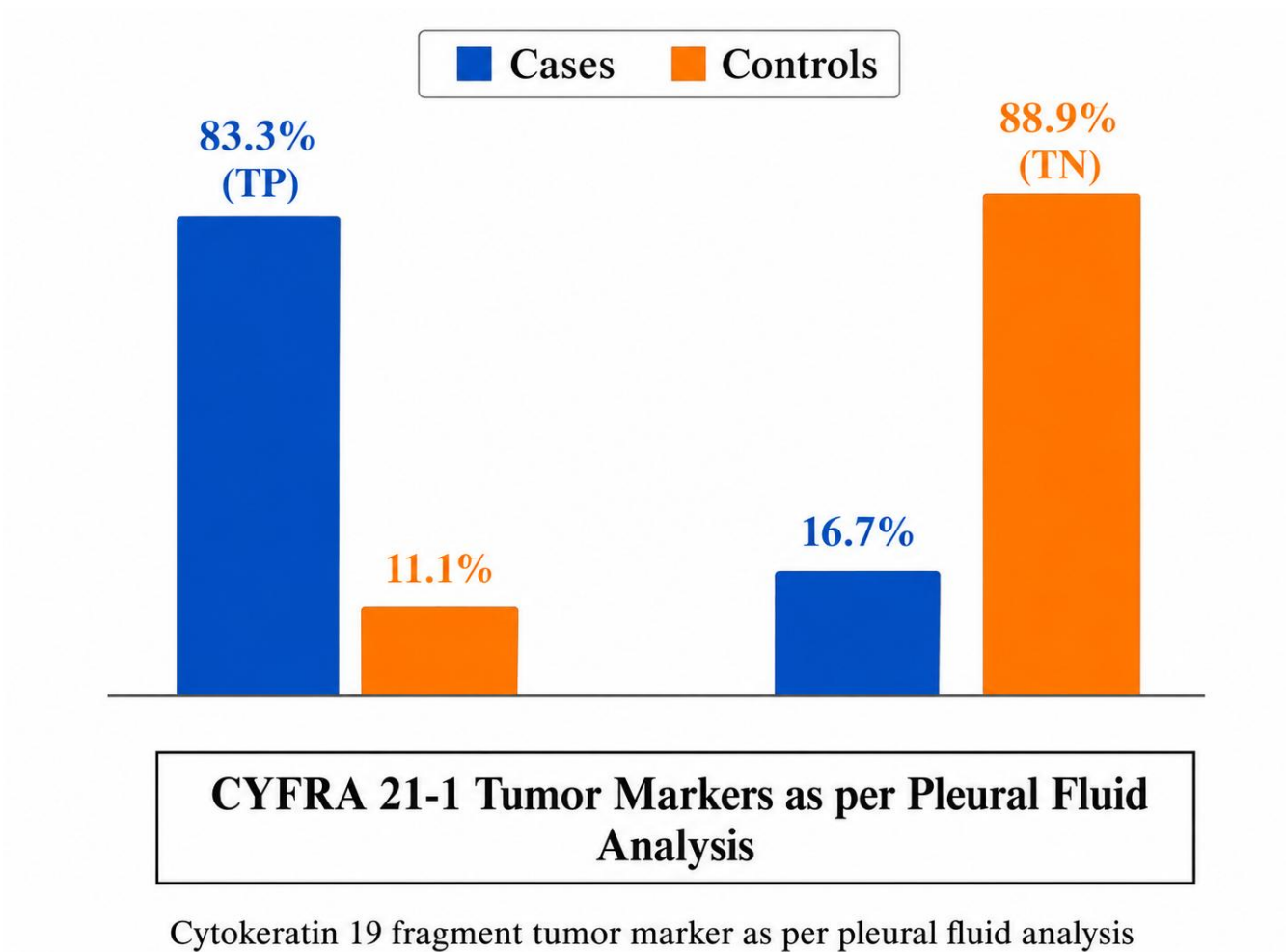
# CYFRA 21-1 as A Biomarker in FNA washout in Other Cancers

Correlation of washout CYFRA 21-1 and FNAC of axillary LNs in breast cancer



Choi JS et al., Fine-needle aspirate CYFRA 21-1, an innovative new marker for diagnosis of axillary lymph node metastasis in breast cancer patients. *Medicine (Baltimore)*. 2015 May;94(19):e811.

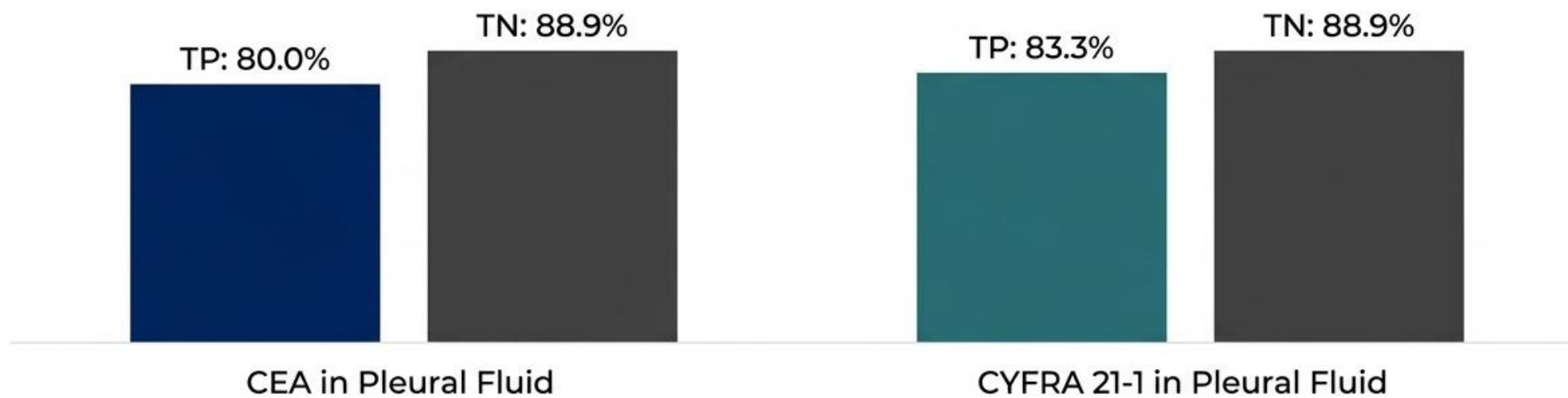
# CYFRA 21-1 as A Biomarker in Pleural Fluid in Other Cancers



Sharma et al., Diagnostic utility of serum and pleural fluid carcinoembryonic antigen, and cytokeratin 19 fragments in patients with effusion from non-small cell lung cancer. *J Carcinog.* 2015 Nov 27;14:7

# Unmatched Sensitivity in Fluid Suspensions

CYFRA 21-1's utility extends perfectly into high-volume fluid environments, outperforming CEA in differentiating malignant from benign pleural effusions in NSCLC.



- When multiplexing Serum + Pleural Fluid CYFRA 21-1, diagnostic sensitivity reached an unprecedented 96.7%.
- Takeaway: Soluble CK-19 fragments maintain remarkable diagnostic integrity in fluid suspensions, laying the groundwork for cyst analysis.

Sharma et al., Diagnostic utility of serum and pleural fluid carcinoembryonic antigen, and cytokeratin 19 fragments in patients with effusion from non-small cell lung cancer. *J Carcinog.* 2015 Nov 27;14:7

# The Thyroid Diagnostic Dilemma: Cystic Lesions

## The Problem



Cystic Papillary Thyroid Carcinoma (cPTC) constitutes 4–13% of all PTCs and presents a unique diagnostic trap. FNAB of these mural nodules often yields large volumes of cyst fluid but insufficient representative tumor cells, tumor cells, routinely resulting in missed malignancy diagnoses.

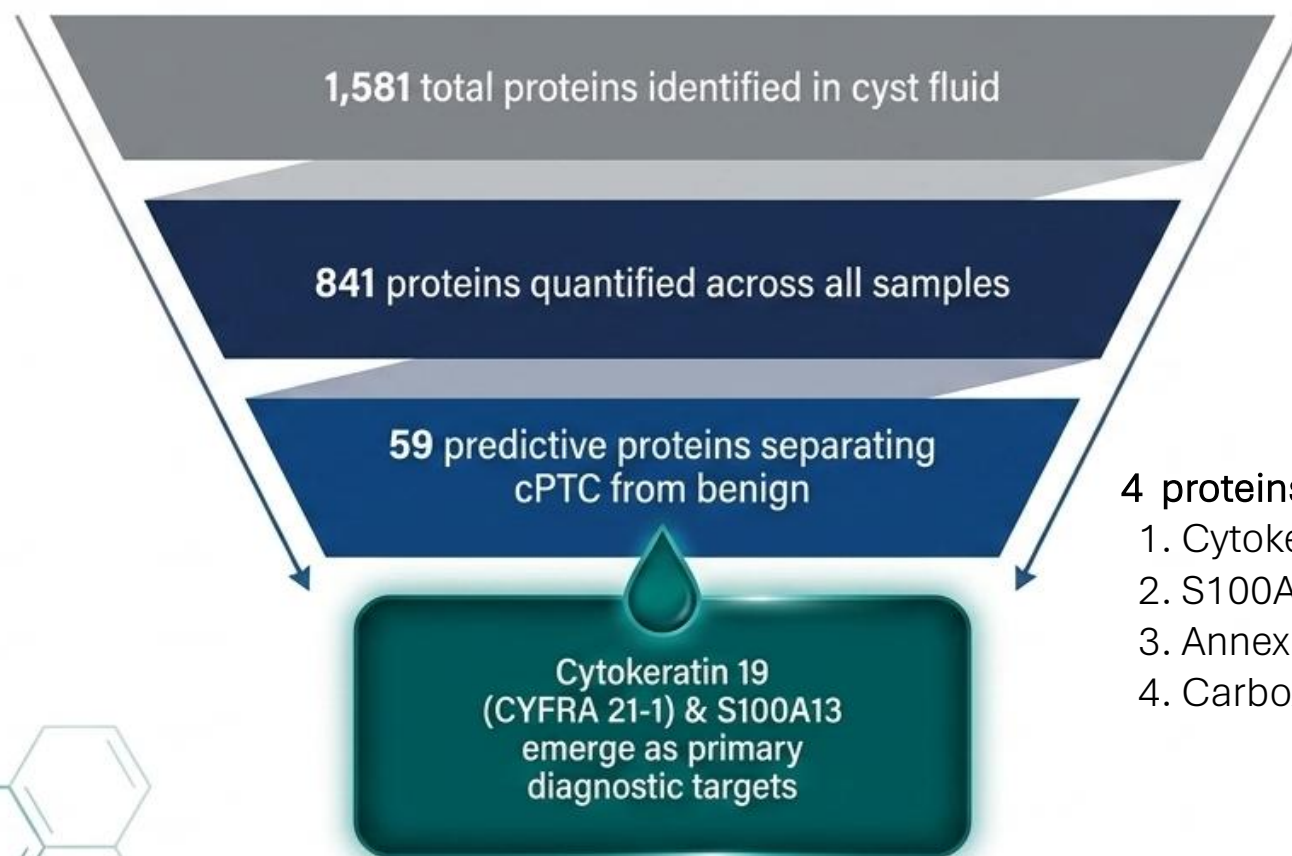
## The Pivot



**If the cellular component is insufficient, could the discarded fluid component hold the diagnostic answer?**

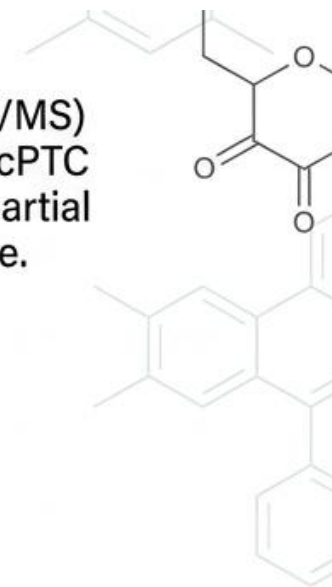
# Uncovering the cPTC Proteome

Using liquid chromatography tandem mass spectrometry (LC-MS/MS) on cyst fluid, researchers profiled the exact molecular makeup of cPTC versus benign lesions. Out of over 1,500 proteins, an orthogonal partial least squares (OPLS) model identified a highly predictive signature.



4 proteins found up-regulated in cystic PTCs

1. Cytokeratin 19 (CK-19)
2. S100A13
3. Annexin A3 (ANXA3)
4. Carboxymethylenebutenolidase (CMBL)

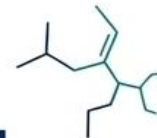
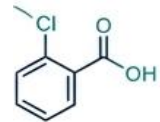
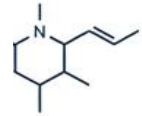
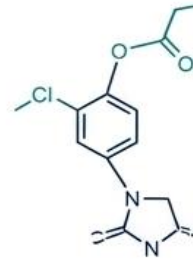
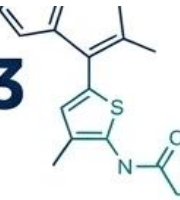


## Selecting the Optimal Marker: CYFRA vs. S100A13

The proteomic screen also identified S100A13 as highly upregulated in cPTC. However, clinical viability requires a balance of sensitivity and specificity to avoid false positives.

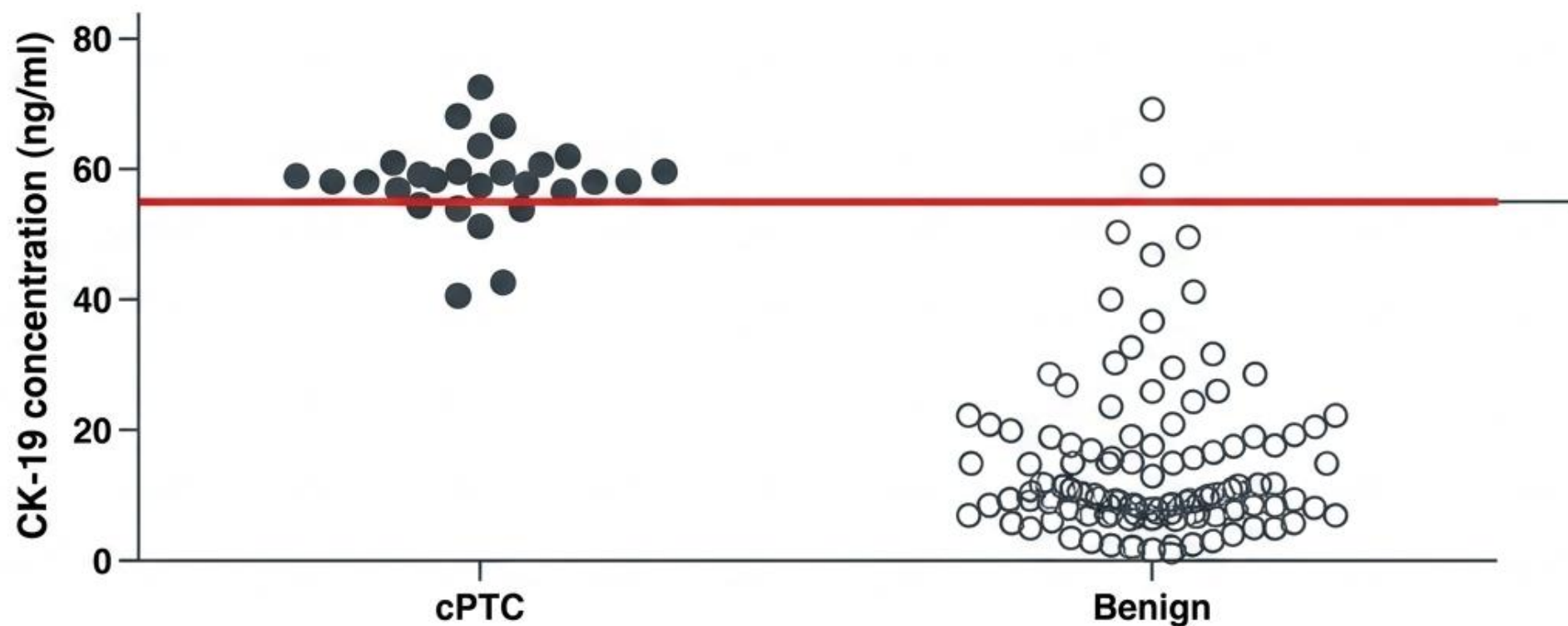
|                           | Sensitivity | Specificity |
|---------------------------|-------------|-------------|
| S100A13<br>(>230 pg/ml)   | 94%         | 35%         |
| CYFRA 21-1<br>(>55 ng/ml) | 82%         | 82%         |

**Takeaway:** While S100A13 suffers from a fatal specificity rate, **CYFRA 21-1 remains the most balanced, reliable, and clinically actionable fluid marker.**



# Establishing the Clinical Threshold: >55 ng/ml

Using standard, widely available ELISA technology (the CYFRA 21-1 assay) on non-depleted cyst fluid, researchers translated the mass spectrometry data into a practical clinical tool.



**The Data:** A cyst fluid concentration of >55 ng/ml for CK-19 yields **82% sensitivity and 82% specificity** for diagnosing cPTC.

## Why did we choose CYFRA 21-1 as a washout biomarker in LN metastasis of thyroid cancer?

- FNAC + washout CYFRA in axillary LN metastasis of Breast Cancer – good diagnostic performance
- Malignant vs. benign in pleural Effusion – diagnostic use in body fluid
- Cystic PTC vs benign cyst – CK 19 as a biomarker in cystic fluid of thyroid cancer
- Mechanistic understanding of CYFRA 21-1 from lung cancer cell-line study

# CYFRA 21-1

- **Cytokeratin 19 (CK19)** : an acidic protein of 40 kDa that is **part of the cytoskeleton of epithelial cells**
  - highly expressed in differentiated thyroid carcinoma, mainly those with papillary histotype
- CYFRA 21-1 : soluble protein fragments derived from CK19 in the circulation of cancer patients
  - ⇒ released by carcinoma cells during intermediate stage of epithelial cell apoptosis and detected by using anti-CYFRA 21-1 antibody

# Methods

- **Study design** : retrospective cohort study
- **Subjects**: 103 patients with thyroid cancer **had both FNACs and washout Tg measurements on 130 suspicious LNs** on thyroid US at that time of initial diagnosis or during cancer surveillance after surgery, and/or during radioiodine remnant ablation **from June 2016 to December 2017**
- **US and FNAC** : performed by experienced radiologist
  - Suspicious LN defined by consensus statement of the Korean Society of Thyroid Radiology
  - LNs with cystic change, calcification, hyperechogenicity, abnormal vascularity, and loss of central hilar echogenicity
  - Samples were sent for cytology along with washout Tg
  - Washout CYFRA was measured on the **remaining samples after FNAC and washout Tg were done**

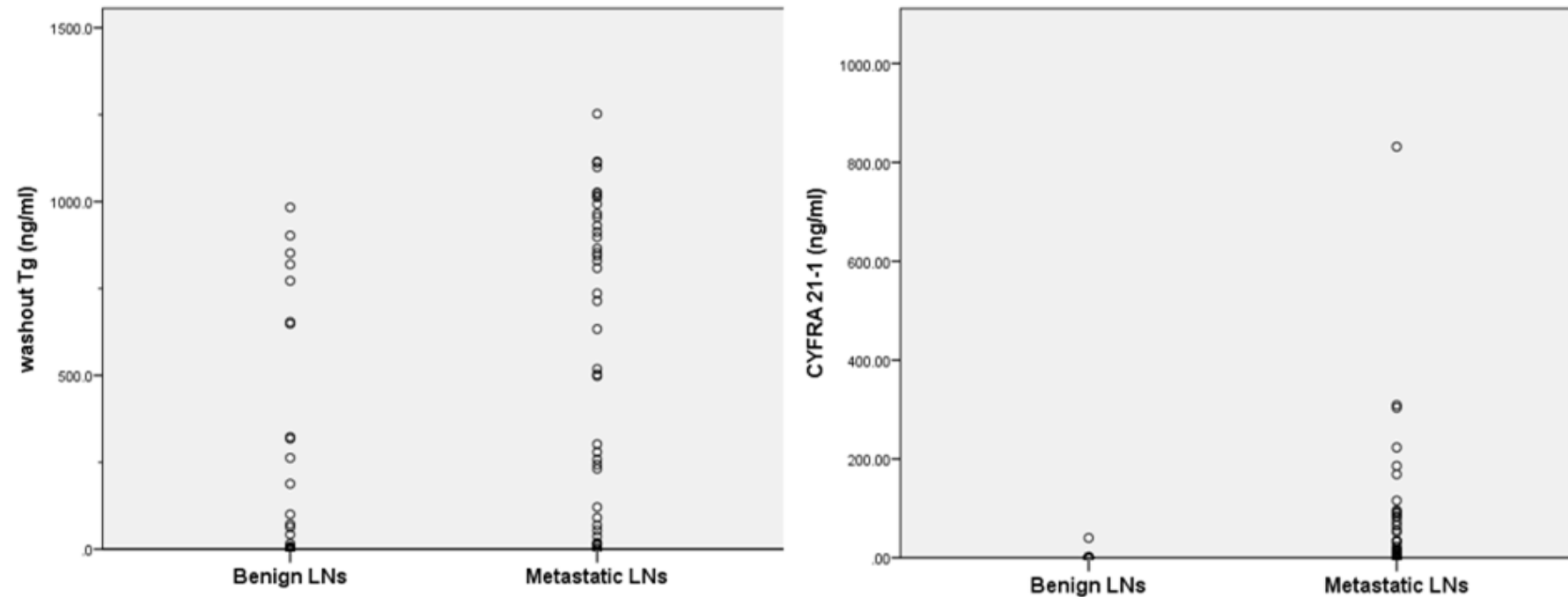
# Methods

- **Final LNs outcomes** : based on postoperative pathologies or FNAC results
- ✓ **Metastatic LNs** : n=42
  - confirmed as metastatic PTC by histopathology results
  - without LNs dissection, **definite metastatic cytologies with high washout Tg levels** and metastatic lesions were identified on neck computed tomography (CT) or magnetic resonance imaging (MRI)
- ✓ **Benign LNs** : - determined from histopathology results of the resected LNs as negative for malignancy
  - without LNs dissection, no increased in size or no interval change LNs during at least one year follow-up thyroid US, CT or MRI finding
  - negative for malignancy after re-biopsy

## Results : Clinical characteristics of 130 Lymph Nodes

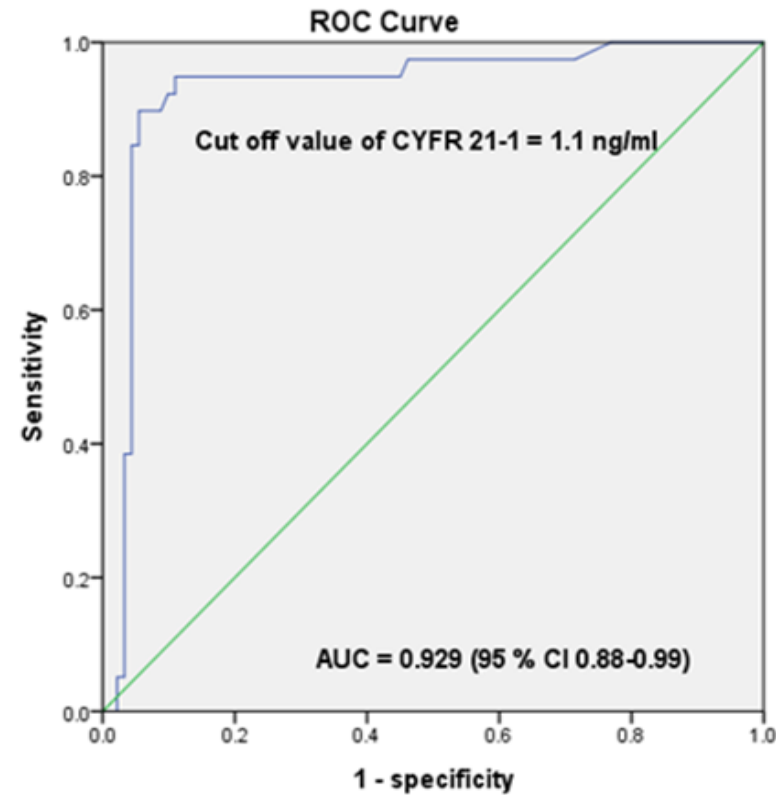
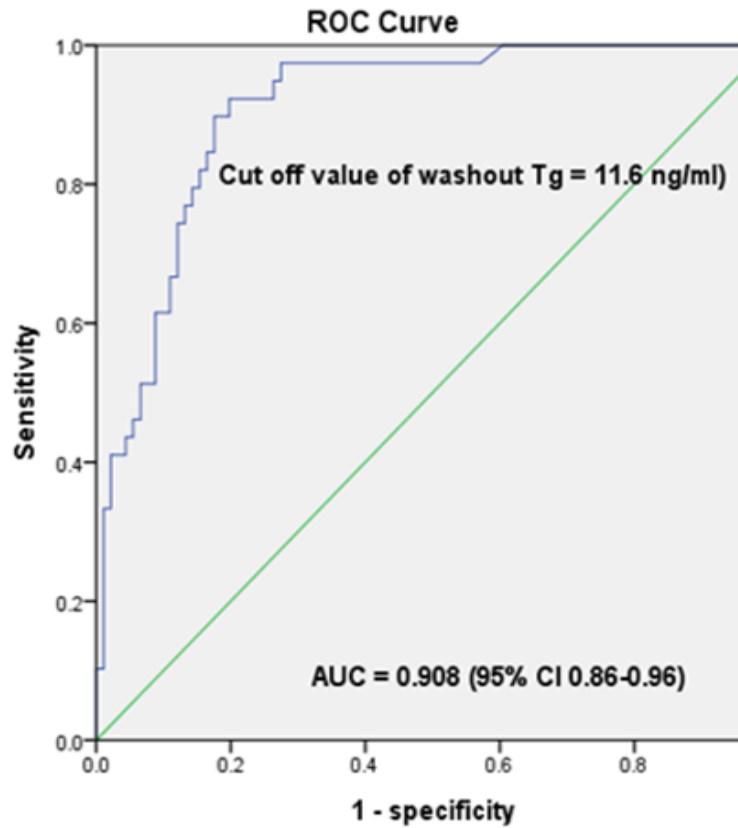
|                                           | Benign LN ( N= 88)         | Metastatic LN (N=42)       | p-value |
|-------------------------------------------|----------------------------|----------------------------|---------|
| FNAC according to operation               |                            |                            | 0.268   |
| FNAC before surgery (%)                   | 37 (42.0)                  | 22 (52.4)                  |         |
| FNAC at postoperative follow-up           | 51 (58.0)                  | 20 (47.6)                  |         |
| Diagnostic methods for final LNs outcomes |                            |                            | 0.000   |
| Histological confirm (%)                  | 42 (47.7)                  | 38 (90.5)                  |         |
| confirmed by image or repeat FNAC (%)     | 46 (52.3)                  | 4 (9.5)                    |         |
| Serum Tg                                  | 15.5 ± 34.1 (0.0-205.0)    | 18.6 ± 27.3 (0.1-126.4)    | 0.614   |
| Serum TgAb                                | 53.12 ± 163.6 (1.6-895.3 ) | 25.8 ± 52.8 (0.1 -270.7)   | 0.319   |
| Washout Tg                                | 80.1± 224.5 (0.0-983.2)    | 568.0 ± 422.6 (0.1-1252.4) | 0.000   |
| Washout CYFRA 21-1                        | 0.9 ± 4.2 (0.1 -40.1)      | 74.2 ± 142.7 (0.2-931.6)   | 0.000   |

## Results : Distribution of washout protein in benign and metastatic LNs



- **Washout Tg and washout CYFRA 21-1 presented higher concentrations in metastatic LNs** than benign LNs, with statistically significance ( $p= 0.000$  and  $p= 0.000$ , respectively).
- While the washout Tg values showed significant overlap between benign and malignant LNs, the CYFRA 21-1 values had little overlap between benign and malignant LNs.

## Results : Cut-off of CYFRA 21-1 and Washout Tg



- Cut off value of **washout Tg** : **11.6 ng/ml** (AUC 0.908 95 % CI 0.86-0.96)
- Cut off value of **CYFRA 21-1** : **1.1 ng/ml** (AUC 0.929 95 % CI 0.88-0.99)

# Results

- Correlation of FNAC, washout Tg and washout CYFRA 21-1 of 130 LNs by final LNs outcomes

| Final diagnosis   | FNAC               |                    | Washout Tg         |                    | Washout CYFRA 21-1 |                    |
|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                   | Positive<br>(N=39) | Negative<br>(N=91) | Positive<br>(N=54) | Negative<br>(N=76) | Positive<br>(N=41) | Negative<br>(N=89) |
| Benign (N=88)     | 3                  | 85                 | 15                 | 73                 | 2                  | 86                 |
| Metastatic (N=42) | 36                 | 6                  | 39                 | 3                  | 39                 | 3                  |

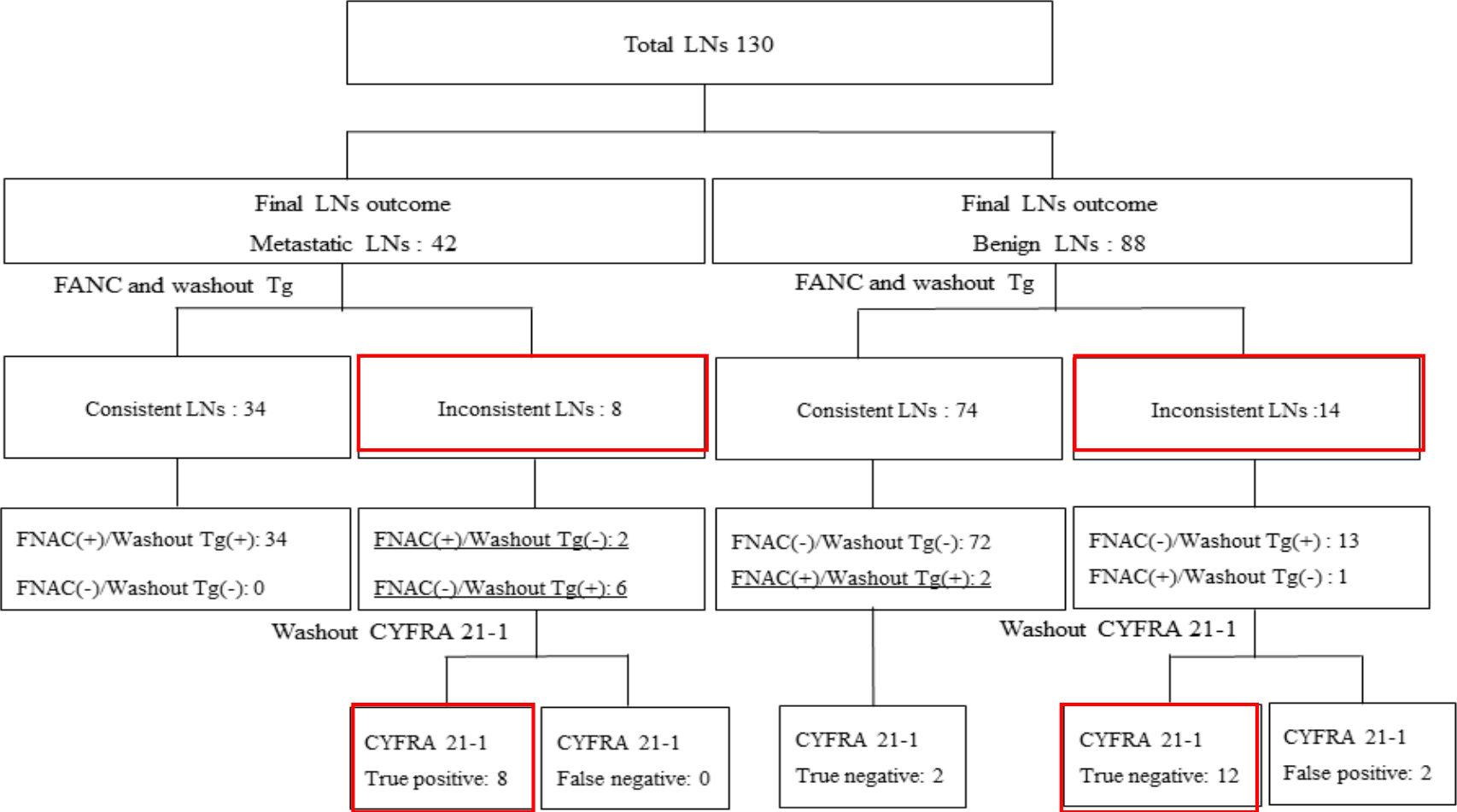
- In 42 metastatic LNs, FNAC missed six, but both washout Tg and washout CYFRA 21-1 missed only three.
- Of the 88 LNs of which the final outcomes were confirmed as benign, washout Tg misclassified 15 as metastatic LNs, while FNAC and washout CYFRA 21-1 misclassified only two and three, respectively.

# Results : Diagnostic performances

- Comparison of diagnostic performance of FNAC, washout Tg, washout CYFRA 21-1, combined with FNAC and washout Tg, combined with FNAC and washout CYFRA 21-1 and combine with all modality

| Diagnostic tool                          | Diagnostic value |                 |         |         |              |
|------------------------------------------|------------------|-----------------|---------|---------|--------------|
|                                          | Sensitivity (%)  | Specificity (%) | PPV (%) | NPV (%) | Accuracy (%) |
| A) FNAC                                  | 85.7             | 96.5            | 93.4    | 91.2    | 93.1         |
| B) Washout Tg                            | 92.9             | 83.0            | 72.2    | 96.1    | 86.2         |
| C) FNAC+ washout Tg                      | 88.1             | 83.0            | 92.5    | 81.1    | 82.3         |
| D) Washout CYFRA 21-1                    | 92.9             | 97.7            | 95.1    | 96.7    | 96.2         |
| E) FNAC+ washout CYFRA 21-1              | 91.9             | 96.5            | 81.6    | 98.8    | 94.2         |
| F) FNAC+ washout Tg + washout CYFRA 21-1 | 98.8             | 93.1            | 98.8    | 88.2    | 90.0         |

# Results : Diagnostic Performance in discordant results



- 22 (16.9%) of the 130 LNs could be confused by FNAC and washout Tg. By applying washout CYFRA 21-1 criteria, 90.9% (20/22) of patients with incorrect diagnoses could be rescued as an accurate diagnosis

## Results : When this washout CYFRA 21-1 is useful?

- Washout CYFRA 21-1 can improve the diagnostic accuracy of FNAC and washout Tg in metastatic LNs of thyroid cancer, regardless of thyroid surgery status.
  - the preoperative cut-off value for washout CYFRA 21-1 was 0.85 ng/mL  
(AUC 0.965, sensitivity 90.9%, and specificity 99.5%, 95% CI 0.904.-0.990)
  - the postoperative cutoff value of washout CYFRA 21-1 was 0.98 ng/mL  
(AUC 0.967, sensitivity 95%, and specificity 98%, 95% CI 0.918.-0.999)
- Washout CYFRA 21-1 could be evaluated when FNAC and washout Tg show discordant results.

# Summary and Conclusions

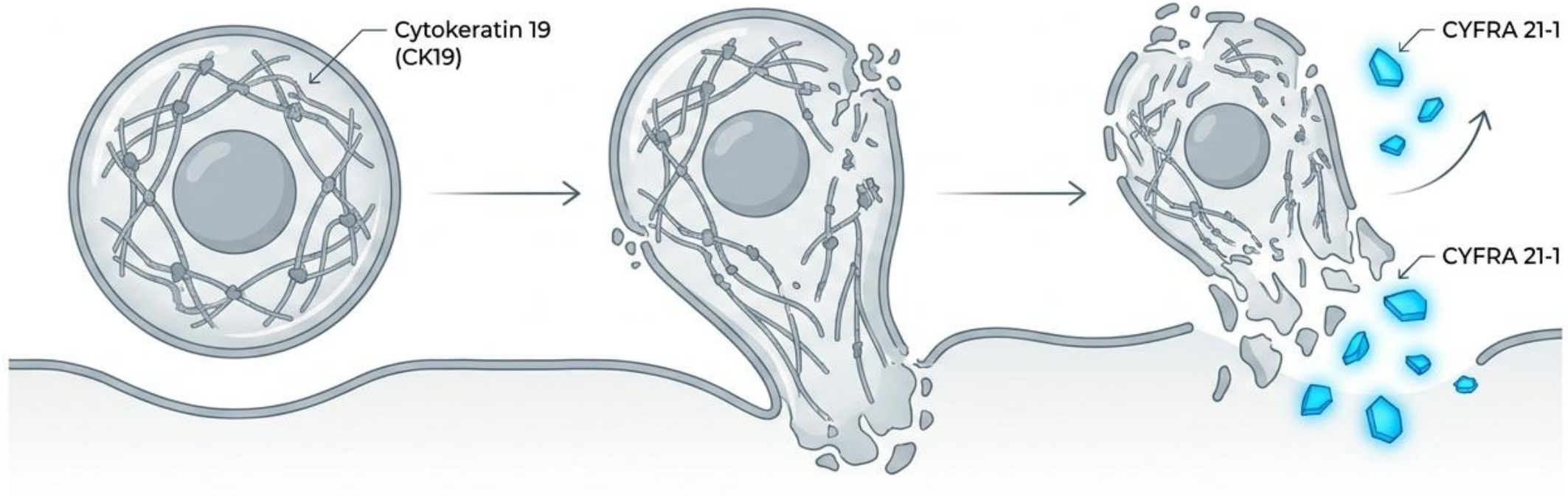
- We investigate differing washout CYFRA 21-1 concentration between benign LNs and metastatic LNs.
- To the best of our knowledge, washout CYFRA 21-1 has not been investigated in LNs of thyroid cancer.
- Washout CYFRA 21-1 lead to improved overall sensitivity and specificity of FNAC and washout Tg for metastatic LNs.
- Further studies will be necessary to validate our preliminary results; larger prospective studies are needed.

# Enter Washout CYFRA 21-1

STEP 1: TUMOR CELL & CK19 SCAFFOLDING

STEP 2: EMT & FRAGMENTATION

STEP 3: CYFRA 21-1 RELEASE



The Biological Mechanism: CYFRA 21-1 is a soluble fragment released during the apoptosis and breakdown of Cytokeratin 19 (CK19) as aggressive tumor cells invade the lymph nodes.

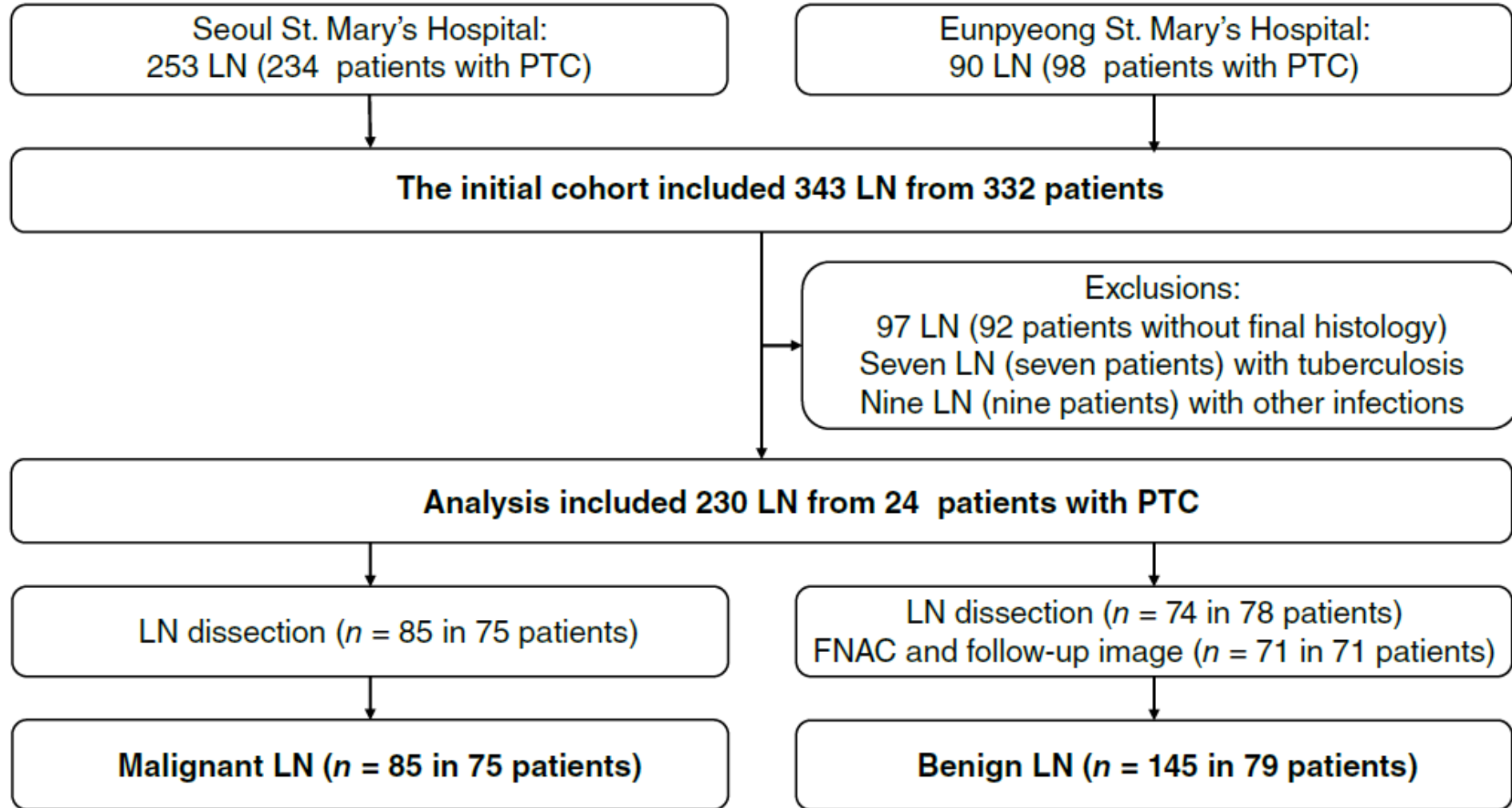
# Reject of Preliminary data (washout CYFRA 21-1) and then...

## DECISION LETTER

### Associate Editors' Comments to Author:

The concept and scientific question are timely. The pilot data are tantalizing. Please replicate your observations in an independent series and should the two datasets replicate, I will send your new manuscript out for review.

# Flow of the Extension Study (Validation Study)



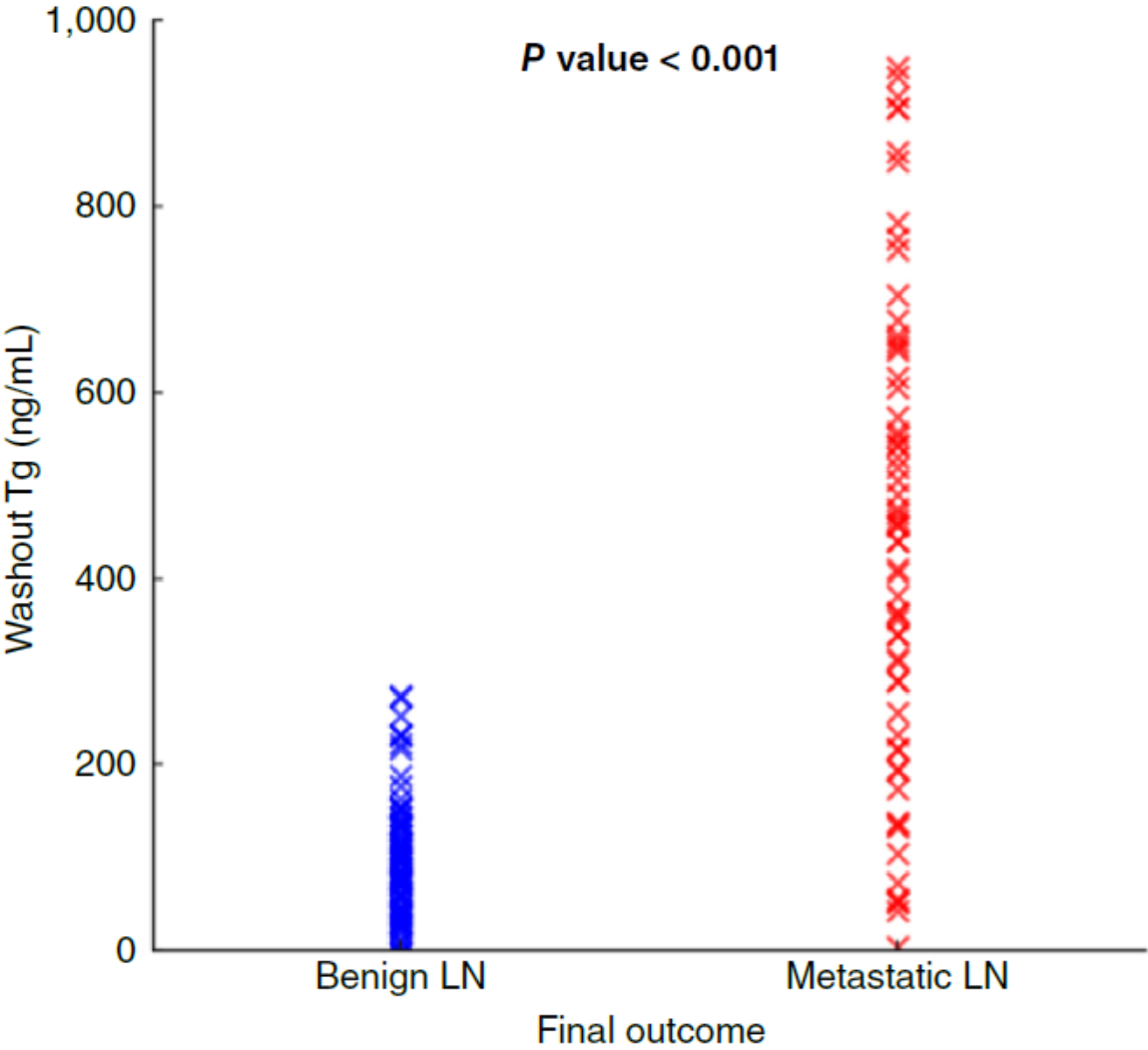
# Baseline Characteristics According to the Final Outcome

| Clinical parameter        | Benign LN (n = 145) | Metastatic LN (n = 85) | P value |
|---------------------------|---------------------|------------------------|---------|
| Age, years                | 56.29 ± 14.83       | 56.16 ± 15.35          | 0.416   |
| Sex, male/female, n (%)   | 58 (40)/87(60)      | 34(40)/51(60)          | 0.394   |
| FNAC time point, n (%)    |                     |                        | <0.001  |
| Preoperative              | 77 (53.1)           | 40 (47.1)              |         |
| Follow-up                 | 68 (46.9)           | 45 (52.9)              |         |
| Serum Tg, ng/mL           | 21.56 ± 57.09       | 95.44 ± 171.39         | <0.001  |
| Serum Tg antibody, IU/mL  | 16.35 ± 23.55       | 29.39 ± 116.90         | 0.065   |
| Washout Tg, ng/mL         | 22.15 ± 110.21      | 495.29 ± 494.63        | <0.001  |
| Washout CYFRA 21-1, ng/mL | 0.67 ± 0.38         | 58.88 ± 216.02         | <0.001  |

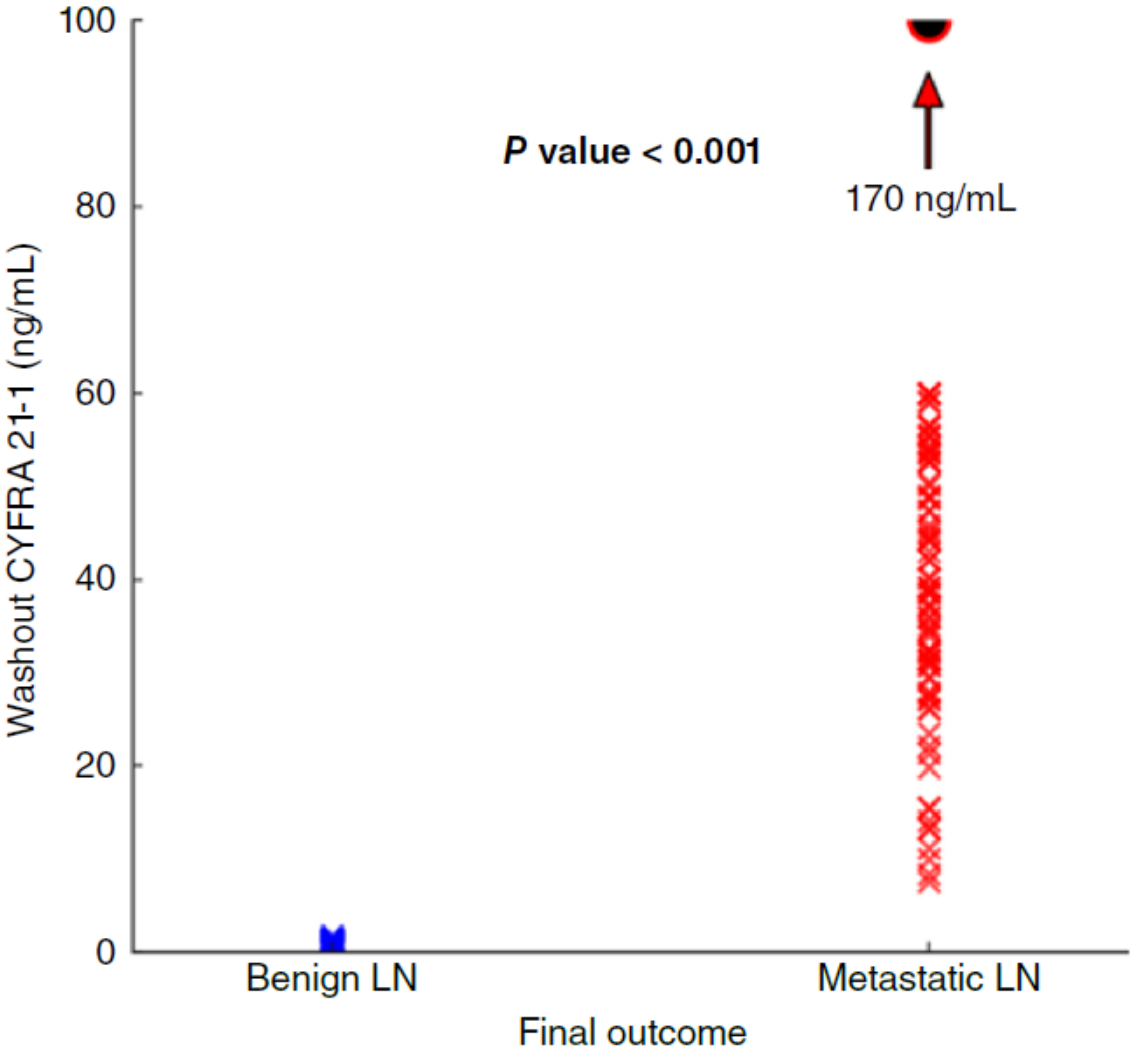
LN, lymph node; PTC, papillary thyroid carcinoma; FNAC, fine needle aspiration cytology

# Flow of the Extension Study (Validation Study)

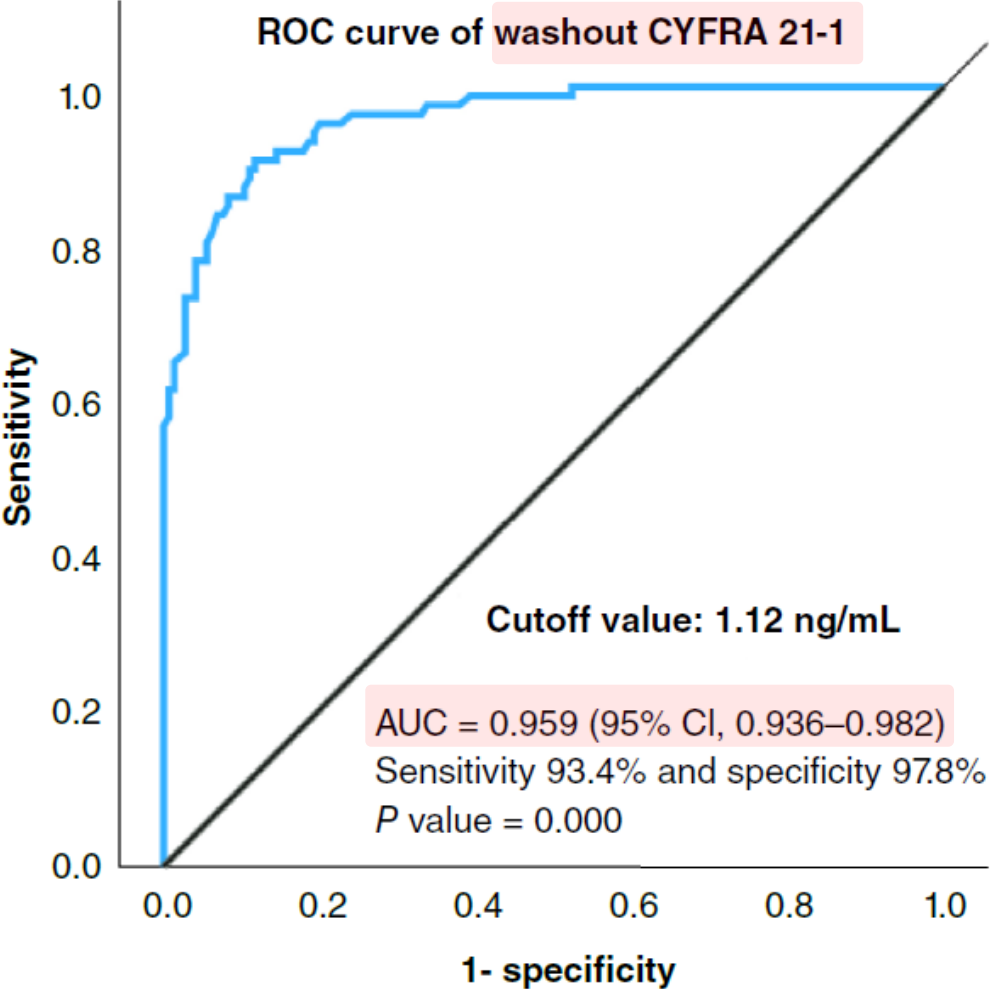
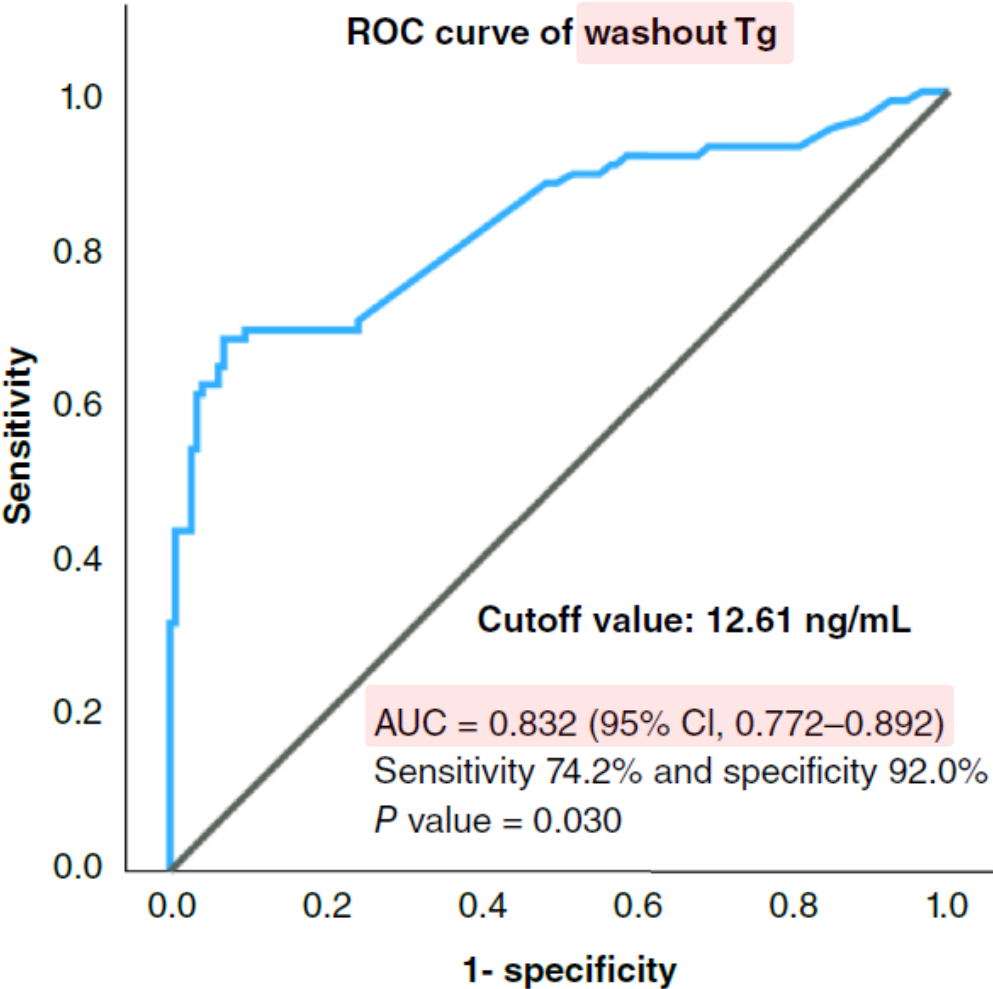
Distribution of washout Tg values in benign and metastatic LN



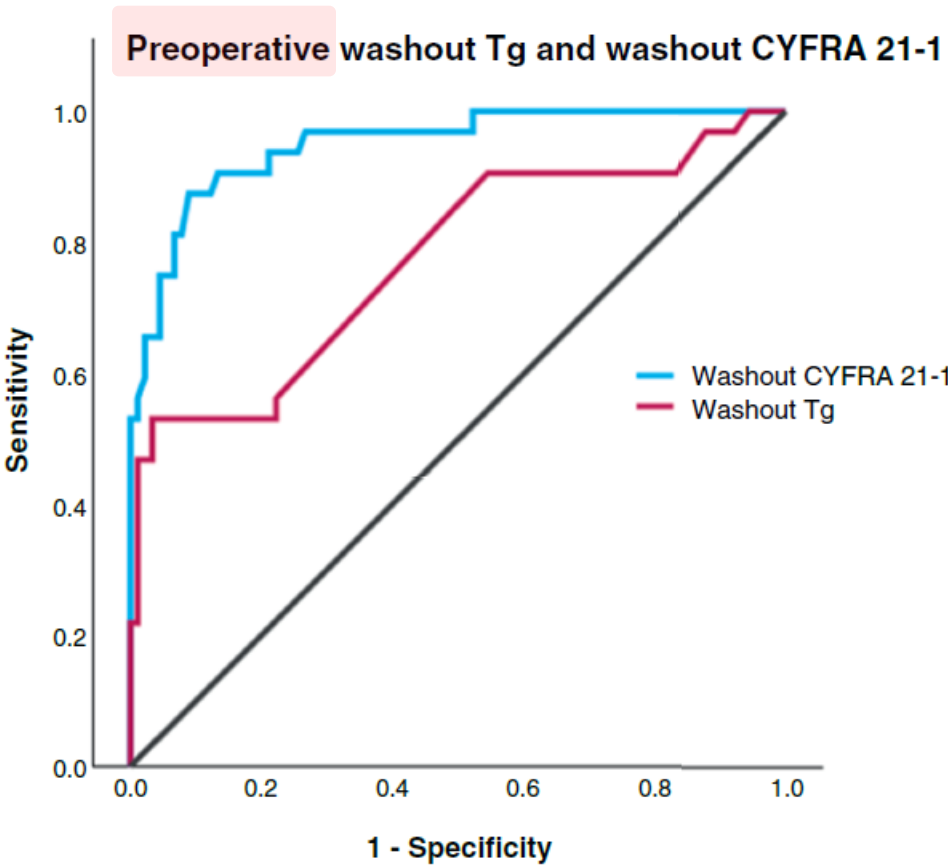
Distribution of washout CYFRA 21-1 values in benign and metastatic LN



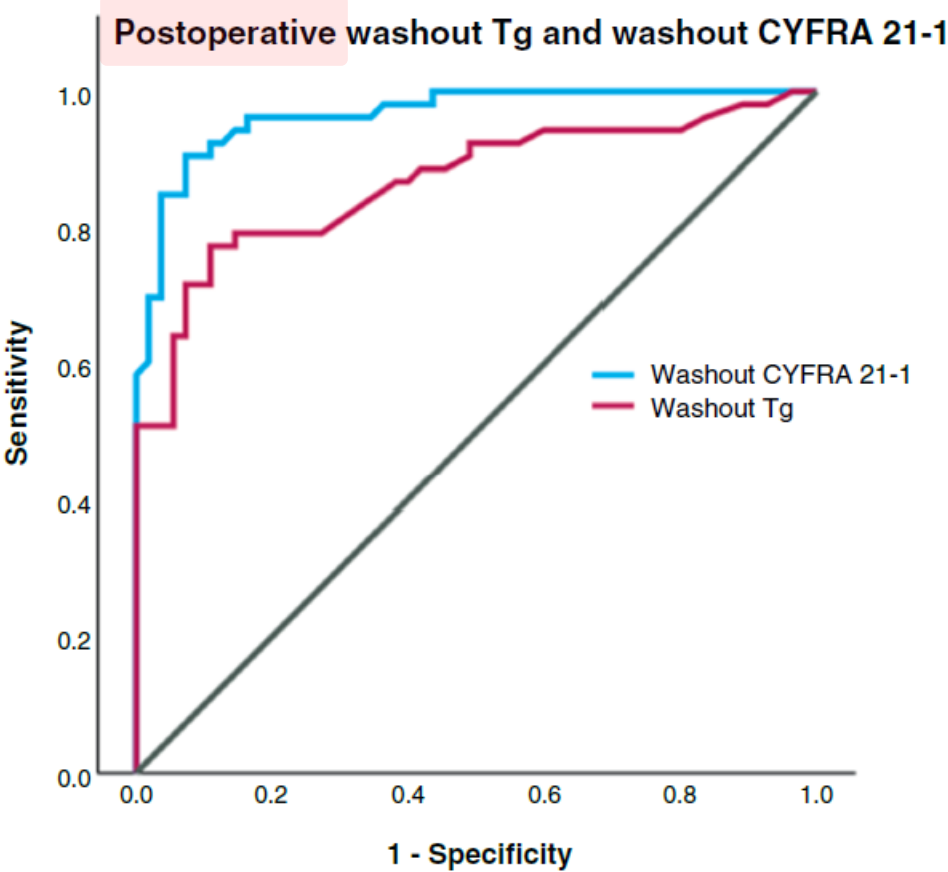
# Cutoff Value of Washout Tg and Washout CYFRA 21-1 By ROC Curve Analysis



# Cutoff Value of Washout Tg and Washout CYFRA 21-1 By ROC Curve Analysis

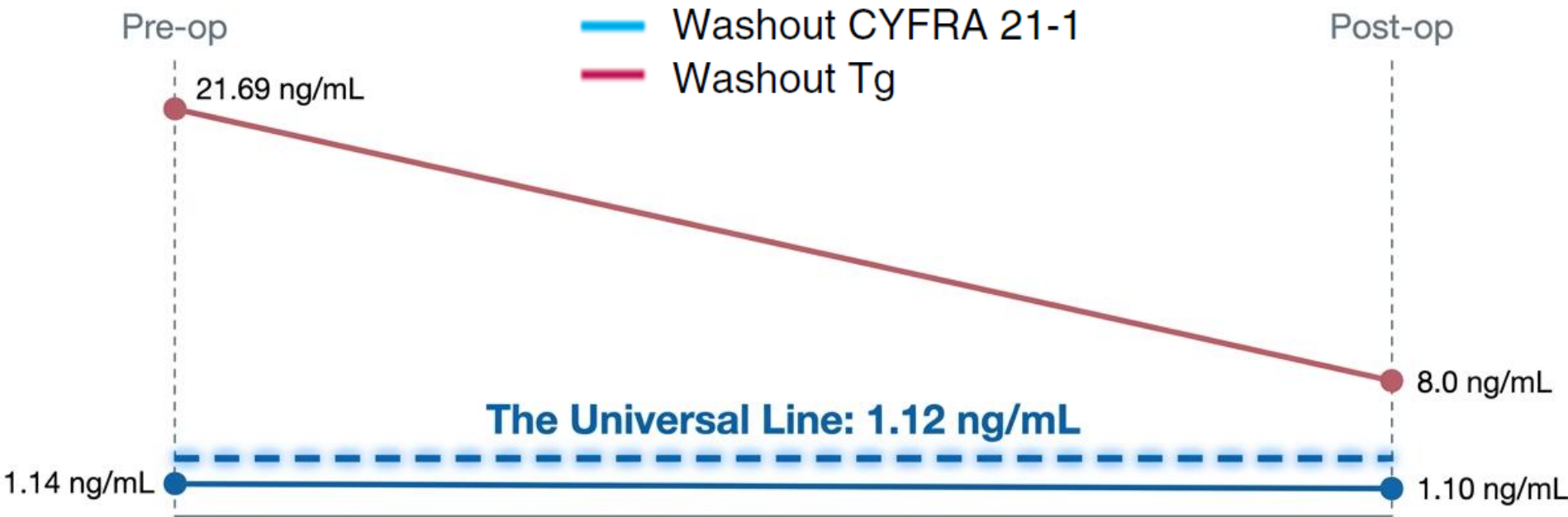


- **Cut off value of washout Tg: 21.69 ng/mL**  
AUC = 0.774 (95% CI 0.670-0.877), Sensitivity 72.9%, Specificity 89.9%  
P-value=0.000
- **Cut off value of washout CYFRA 21-1: 1.14 ng/mL**  
AUC = 0.949 (95% CI 0.908-0.990), Sensitivity 90.6%, Specificity 94.5%  
P-value=0.000



- **Cut off value of washout Tg: 8.0 ng/mL**  
AUC = 0.867 (95% CI 0.797-0.938), Sensitivity 77.4%, Specificity 89.1%  
P-value=0.000
- **Cut off value of washout CYFRA 21-1: 1.10 ng/mL**  
AUC = 0.966 (95% CI 0.937-0.995), Sensitivity 94.3%, Specificity 98.2%  
P-value=0.000

# Cutoff Value of Washout Tg and Washout CYFRA 21-1 By Surgery Status



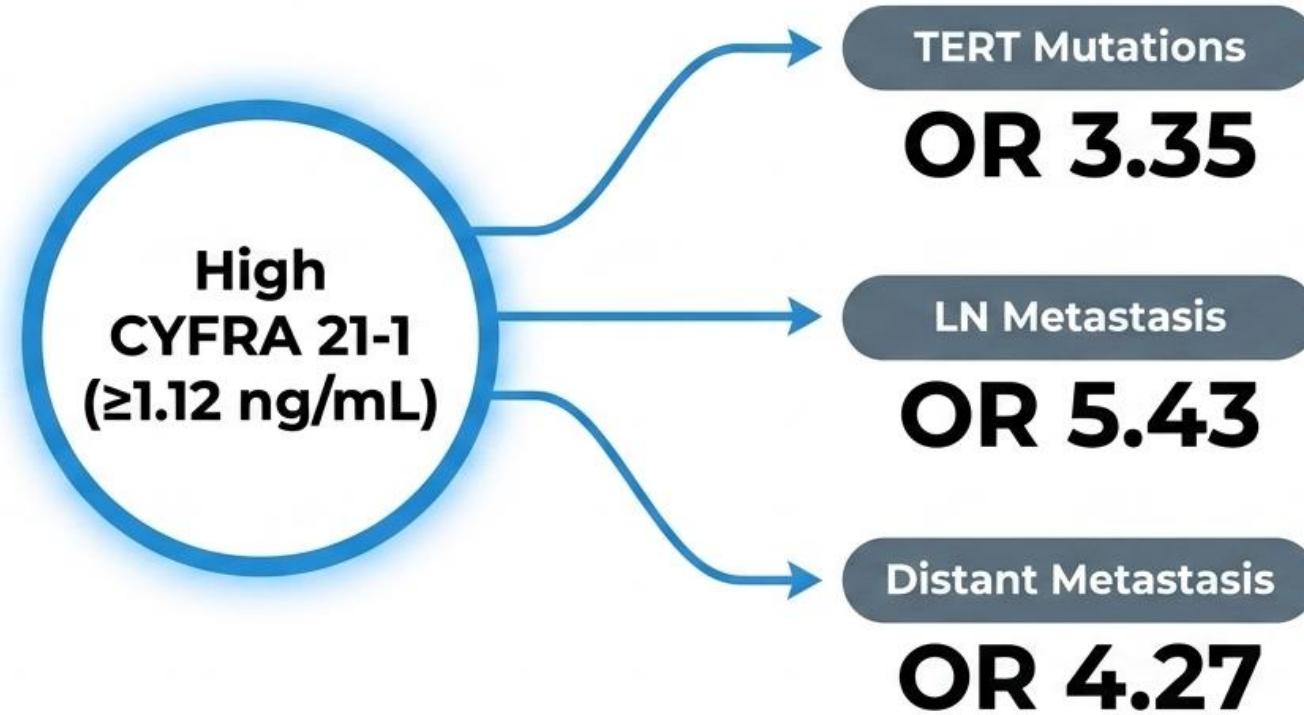
While Washout Tg thresholds wildly fluctuate due to normal thyroid tissue, CYFRA 21-1 maintains a highly stable, universal optimal cutoff of 1.12 ng/mL.

# Head-to-Head Diagnostic Power

| Criteria                    | Washout Tg                | Washout CYFRA 21-1      |
|-----------------------------|---------------------------|-------------------------|
| Sensitivity                 | 74.2%                     | <b>93.4%</b>            |
| Specificity                 | 92.0%                     | <b>97.8%</b>            |
| Affected by Normal Thyroid? | <b>Yes</b> (Interference) | <b>No (Independent)</b> |
| Affected by Tg Antibodies?  | <b>Yes</b> (Interference) | <b>No (Independent)</b> |

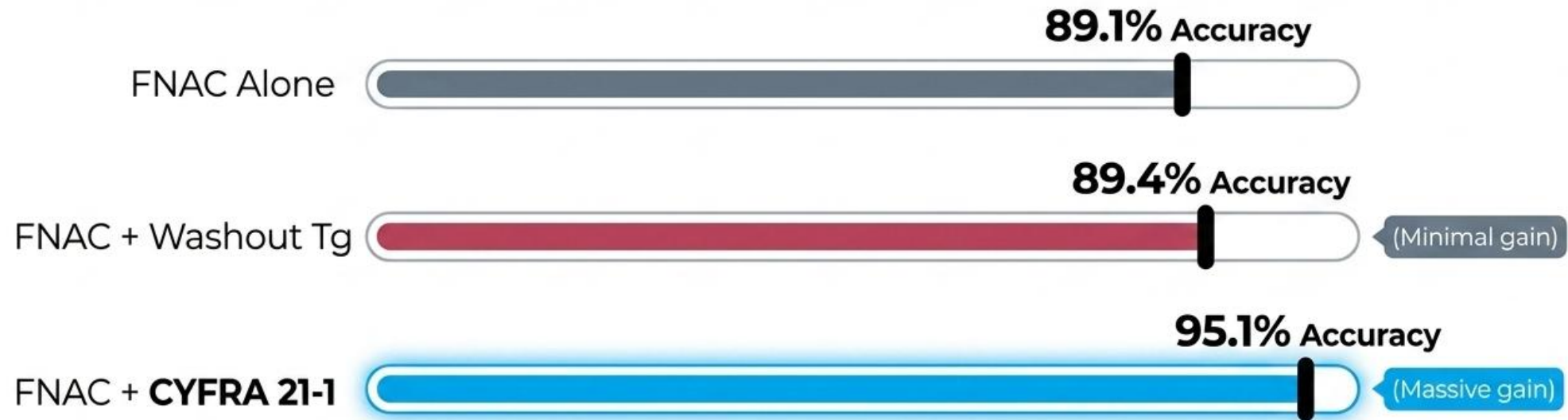
CYFRA 21-1 demonstrates statistically superior diagnostic performance metrics while remaining entirely immune to the biological interferences that compromise Tg.

## Beyond Diagnosis: Prognostic Power



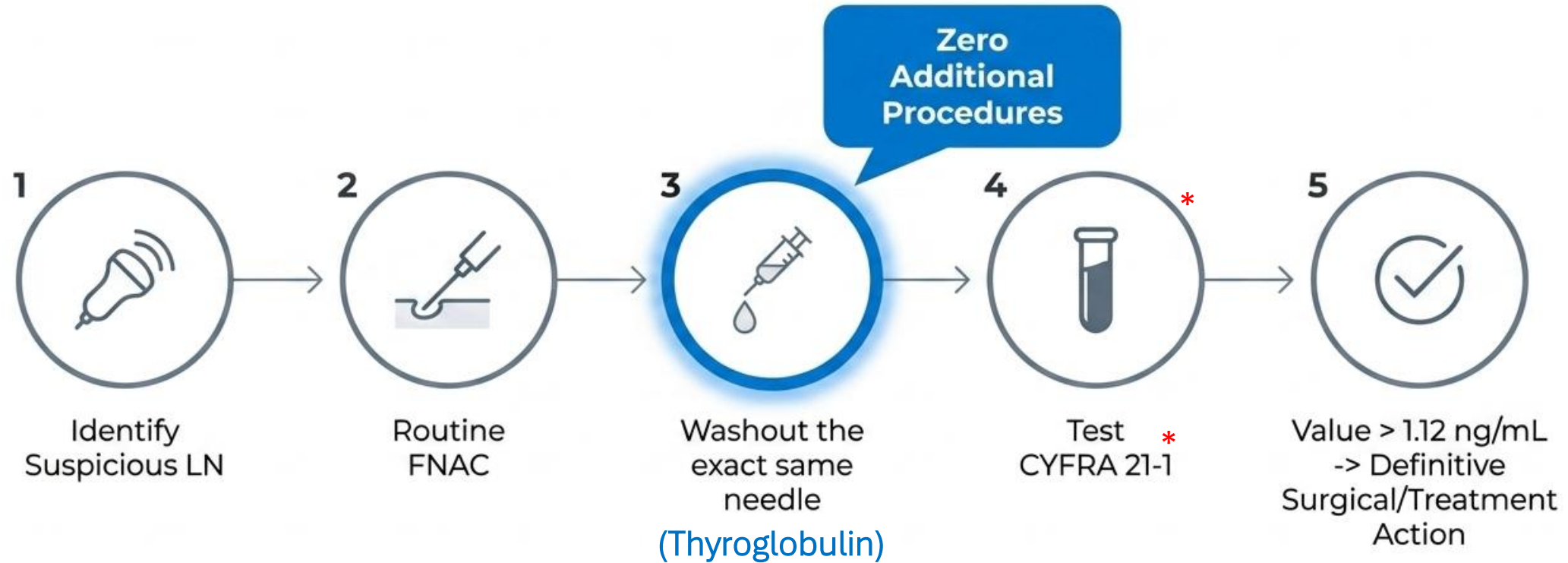
High CYFRA 21-1 levels do more than identify positive nodes; they are intrinsically linked to aggressive tumor behavior, genetic mutations (TERT), and poor long-term prognosis.

# The Ultimate Diagnostic Combination



Combining FNAC with Washout CYFRA 21-1 yields unprecedented diagnostic accuracy. It definitively **minimizes** misdiagnosis, significantly reducing the burden on cytopathologists and **preventing** unnecessary surgeries.

# The New Clinical Paradigm



Seamless Integration. Adopting CYFRA 21-1 as the gold-standard complementary biomarker requires no extra clinical procedures, instantly streamlines diagnosis, and safely prevents unnecessary neck dissections.

# US-guided FNA Washout Biomarkers between Breast Ca and Thyroid Ca

| Category                                  | Breast Cancer                                                                                                                      | Thyroid Cancer                                                                                                                                 |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Main target lymph nodes                   | Axillary lymph nodes                                                                                                               | Cervical lymph nodes<br>(especially recurrent or metastatic nodes)                                                                             |
| Established standard approach             | Axillary US, US-guided FNA or core needle biopsy, sentinel lymph node biopsy (SLNB)                                                | US-guided FNA cytology + washout thyroglobulin                                                                                                 |
| Clinical need for washout biomarkers      | Limited, because effective histologic and surgical staging methods already exist                                                   | High, because cytology and washout Tg can be limited in small, cystic, or postoperative recurrent nodes                                        |
| Marker specificity                        | CYFRA 21-1 is not breast cancer-specific; it reflects epithelial tumor activity and can be elevated in other cancers               | Tg is thyroid-lineage specific; CYFRA 21-1 can serve as a complementary marker, especially in CK19-positive or dedifferentiated thyroid cancer |
| Impact on treatment decisions             | Detection of low-volume nodal disease may not substantially change treatment because axillary surgery is increasingly de-escalated | Accurate nodal diagnosis can directly affect reoperation, RAI strategy, local treatment, or active surveillance decisions                      |
| Recent clinical trend                     | De-escalation of axillary surgery and selective omission of SLNB in low-risk patients                                              | Increasing emphasis on accurate diagnosis of recurrent or metastatic cervical lymph nodes                                                      |
| Practical clinical utility                | Relatively low; mainly a supplemental diagnostic tool rather than a routine test                                                   | Relatively high; may improve diagnostic accuracy when combined with FNAC and washout Tg                                                        |
| Reason for limited or increasing adoption | Limited adoption because it rarely changes management beyond established biopsy/SLNB pathways                                      | Increasing adoption because it addresses real diagnostic gaps in thyroid cancer lymph node evaluation                                          |

SLNB, sentinel LN biopsy; CK19, cytokeratin 19

## TAKE-HOME MESSAGE

- Diagnostic biomarkers - clinical unmet needs / impact on clinical decision
- Sometimes, find well-known biomarkers rather than “fancy” –omics biomarkers  
- only in different situations (treatment status)
- Always keep in mind – molecular characteristics (from production to removal)
- Only one preliminary paper is not enough! Validation process is also important

## On-going/Future Study

- Proteomics of FNA washout fluids for mining other biomarkers
- Cost-effectiveness of washout CYFRA 21-1
- Mechanistic study of LN CYFRA 21-1 – IHC of CK-19 / caspase-3
- Multi-center prospective study
- Application for New Health Technology Decision

# Contributors/Collaborators



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